Join the lecture through the online portal

Newton School of Technology

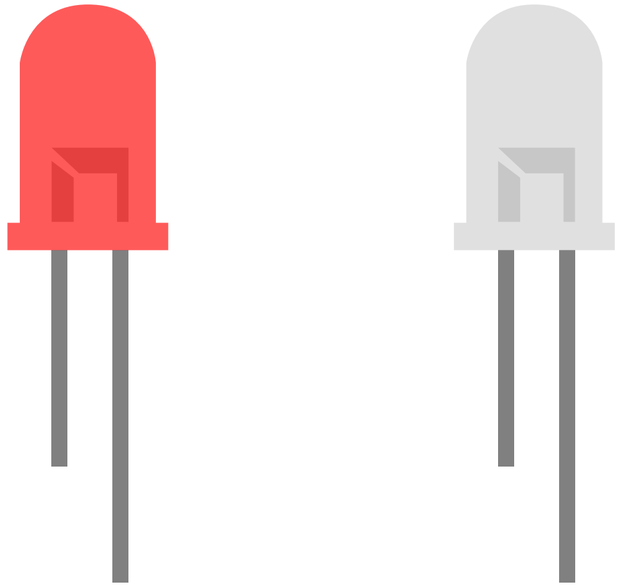
Recap

Electronic Devices

Newton School of Technology

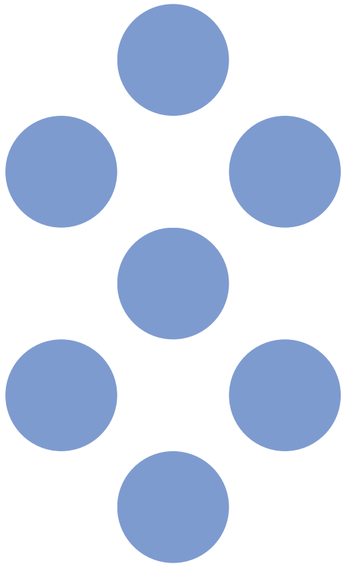
An LED can have two **states**. ON and OFF.

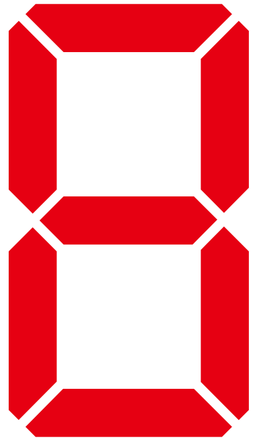
Digital electronic devices work using two electrical states, represented as 0 and 1.



Building a 7-Segment Display

Newton School of Technology

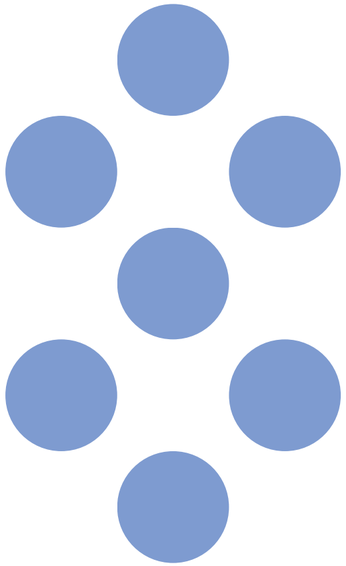


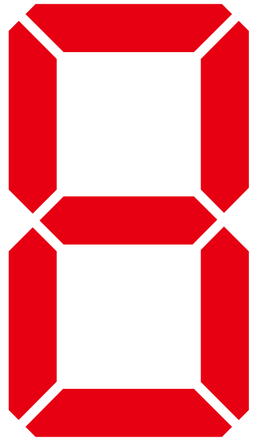


Building a 7-Segment Display

Let us take an example of building a 7-segment display.

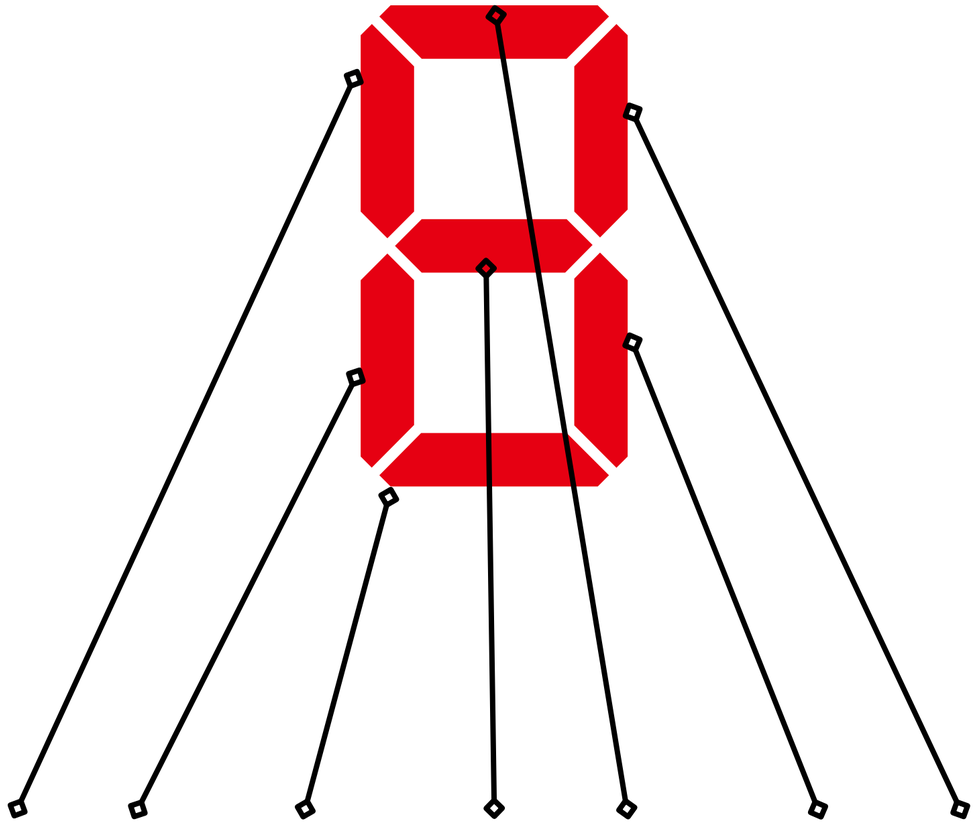
We can arrange 7 LEDs as shown in the diagram to design a 7-segment display.





Building a 7-Segment Display

Each of the individual LED can be connected to a wire and controlled independently.

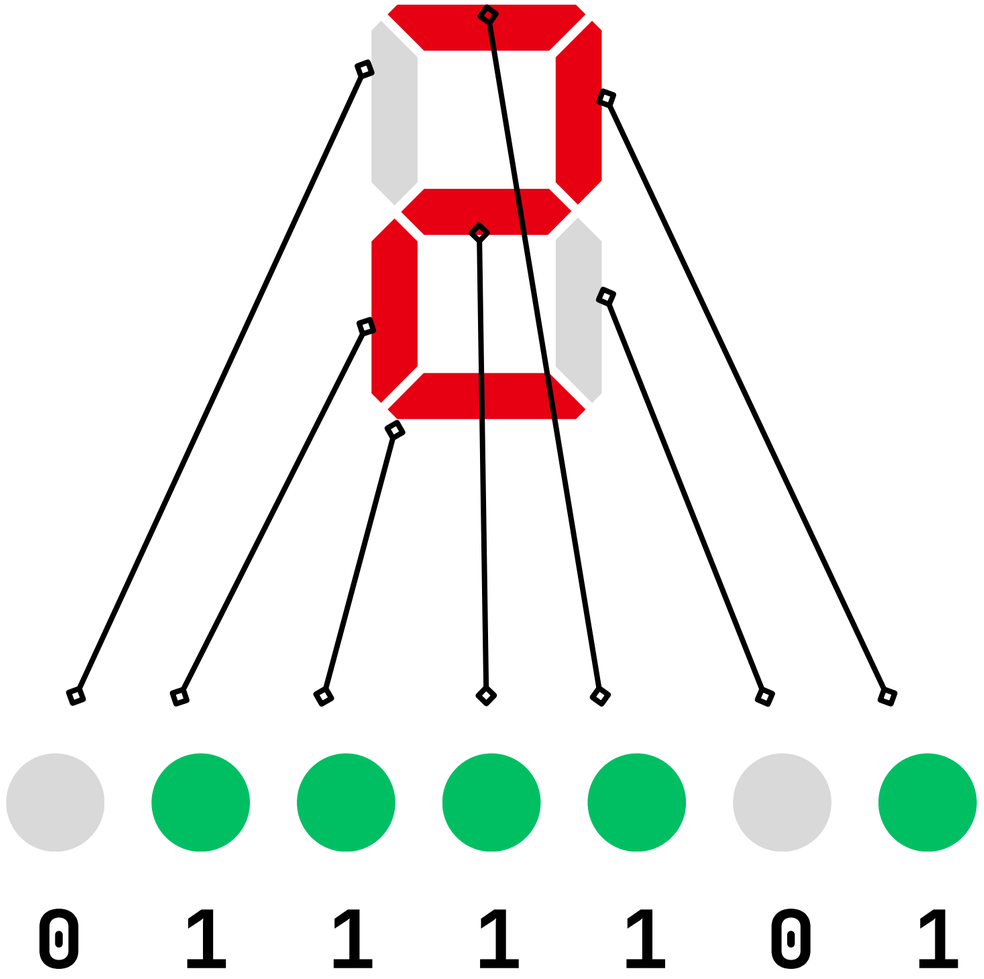


Building a 7-Segment Display

Let’s say we want to display the digit ‘2’.

To do so, we can turn certain LEDs ON and certain LEDs OFF as shown in the figure.

These **states of LEDs can be represented by numbers 0 and 1**.





Newton School of Technology

8

Number Representation and Data Encoding

by

System & AI Essentials

Newton School of Technology

Why Did Humans Need Numbers?

Counting

Humans started counting to **manage resources** and **keep track of important items** for survival. It also emerged naturally from our ability to recognize patterns in the environment and our need to **communicate quantities** precisely.

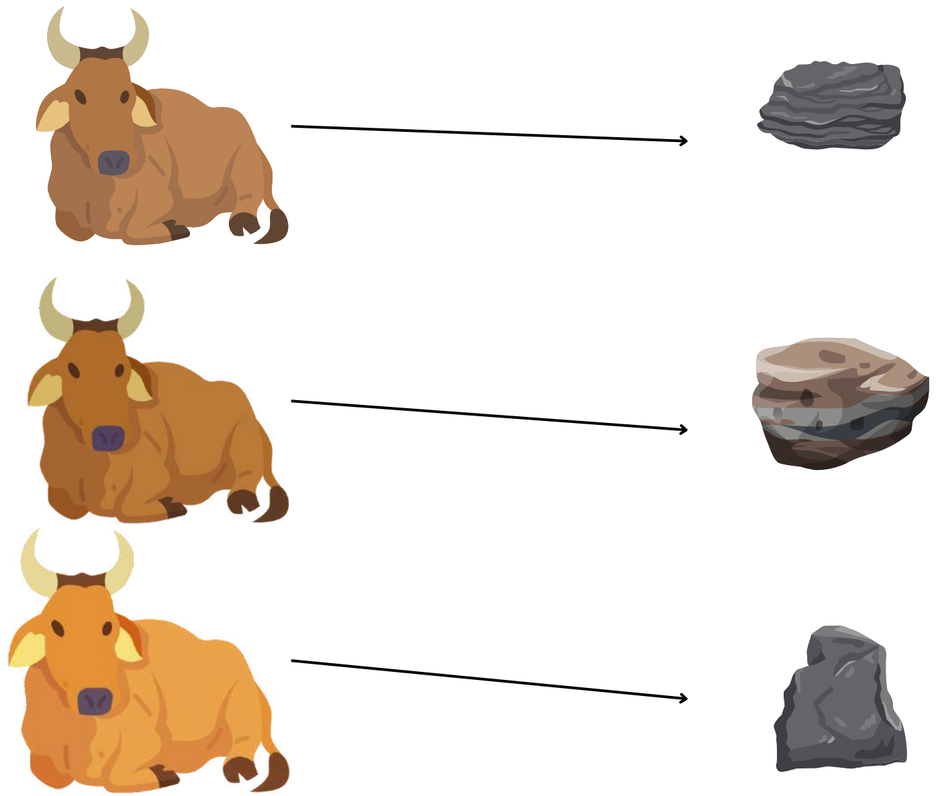




Early humans used counting for:

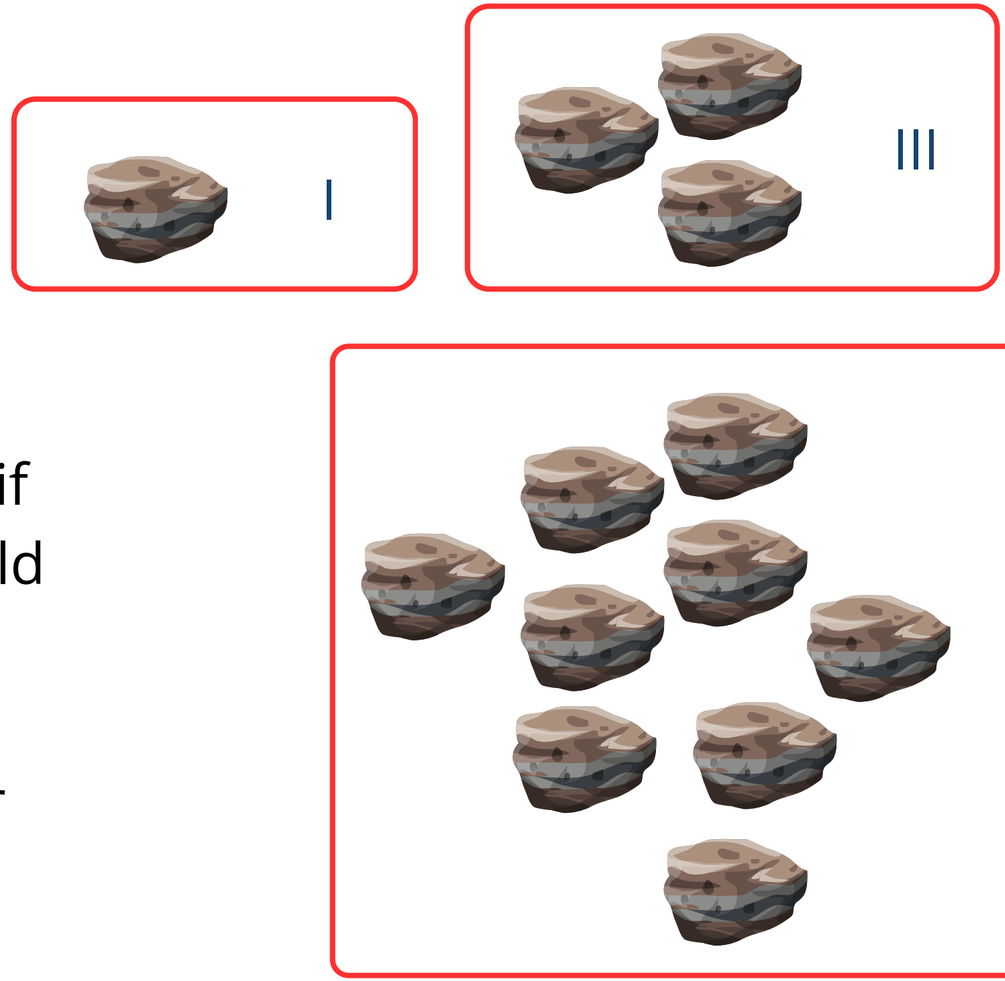
1. Tracking livestock or hunted animals
2. Monitoring supplies
3. Trading goods with other groups
4. Coordinating group activities (e.g., number of days until a hunt)

These basic counting applications helped early humans improve their survival strategies and social organization.



Unary Number System

One Symbol



The most basic form of counting is the **unary number system**, where **each symbol represents one quantity**. So if we want to represent 3 stones, we could use 3 symbols.

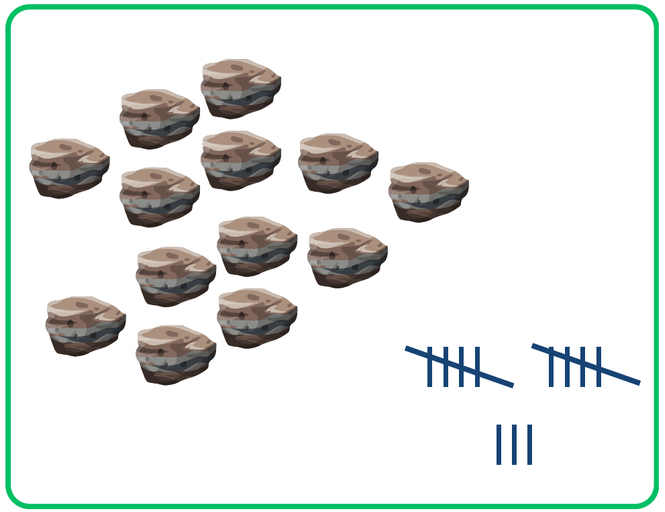
However, it becomes tedious for larger quantities.

Creating New Symbols

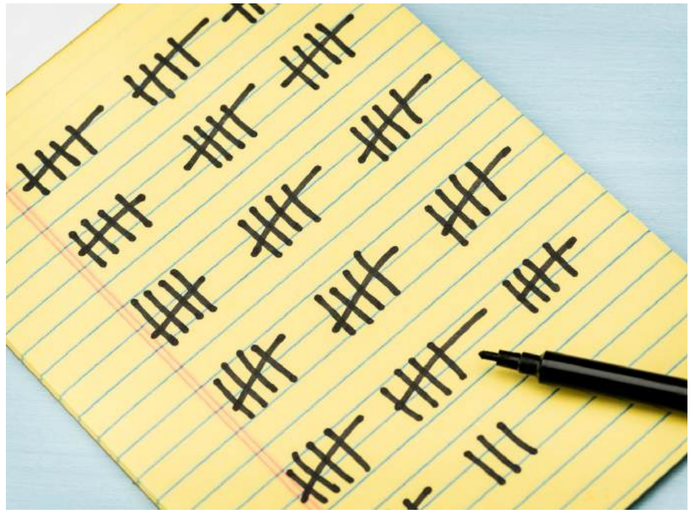
Humans developed more complex counting symbols to represent larger quantities more efficiently.

As you can see, representing a number such as 20 requires 20 symbols in unary number system, but if we decide to use a different symbol for 5, we can represent 20 with just 4 symbols.





Different symbols were created to represent different quantities.



Tally Marks

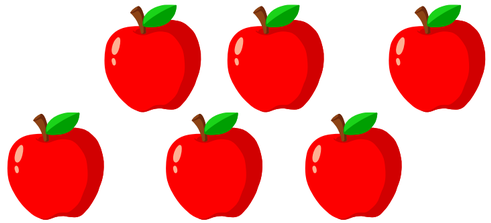
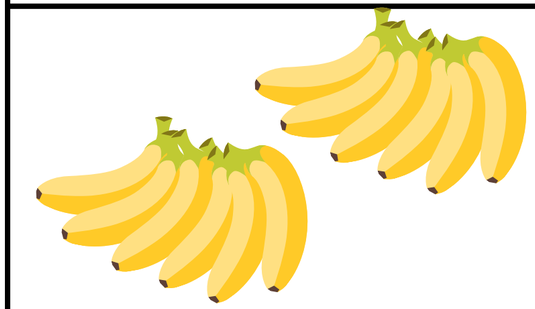
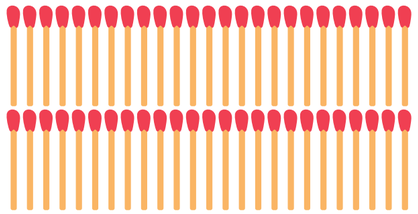


Roman Numbers



Hieroglyphics

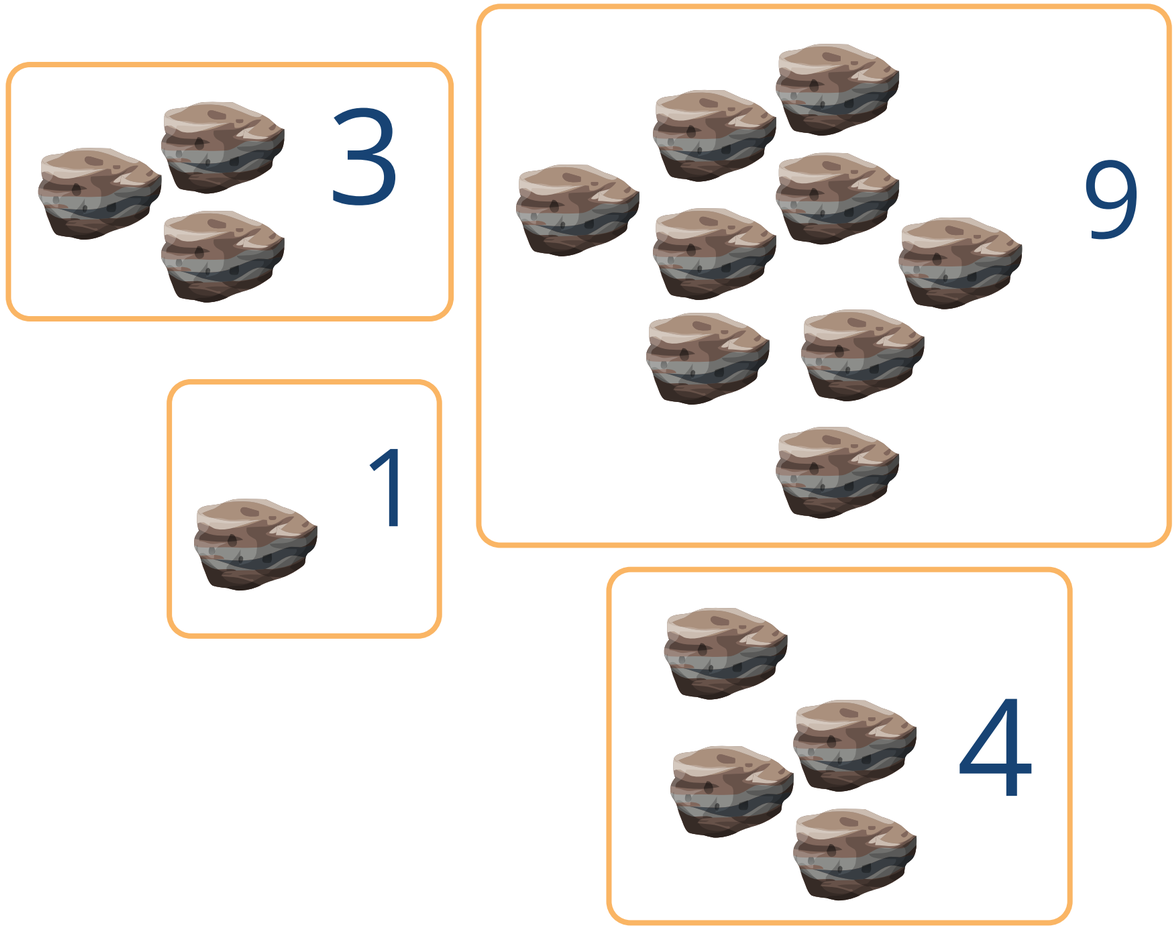
Examples

Items	Unary	Roman	Decimal
	||||||	VI	6
	||||||||||||	XII	12
	|||||||||||\

Decimal Number System

Ten Symbols

The decimal number system, also known as **base-10**, uses **ten digits (0-9)** to represent all numbers.



How We Count

0	10	410	2410
1	11	.	.
2	12	.	.
3	13	.	5172
4	14	512	.
5	15	.	
6	16	.	.
7	17	698	.
8	18	.	8698
9	19	.	.

What is base of a number system?

Newton School of Technology

$$2378_{10}$$

Base

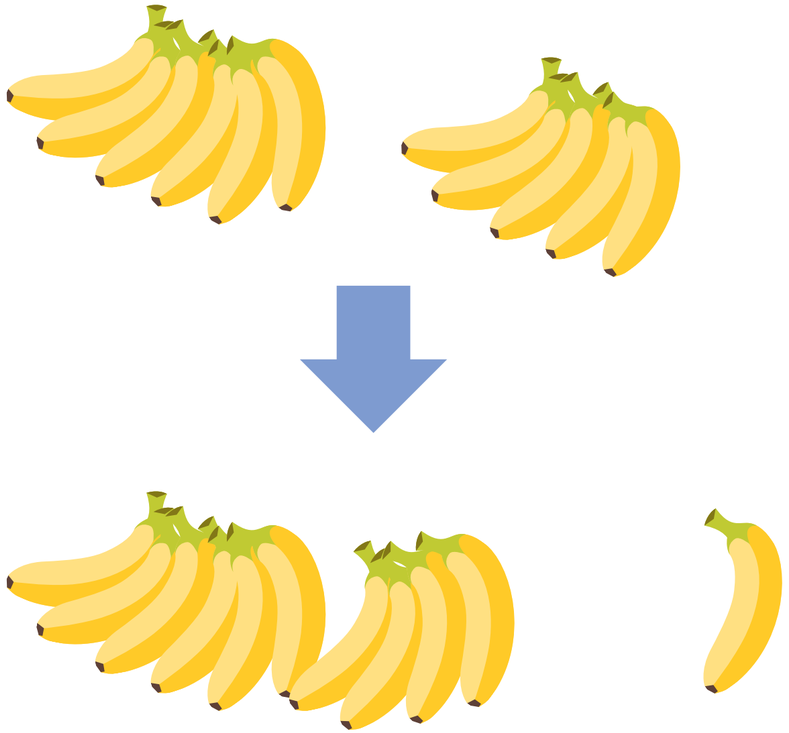
The base of a number system is the **number of unique symbols** used to represent numbers in that system.

Let’s say we have 11 items.
We count a set of 10 and 1 single items.

What about 25 items?
We count 2 sets of 10 and 5 singles.

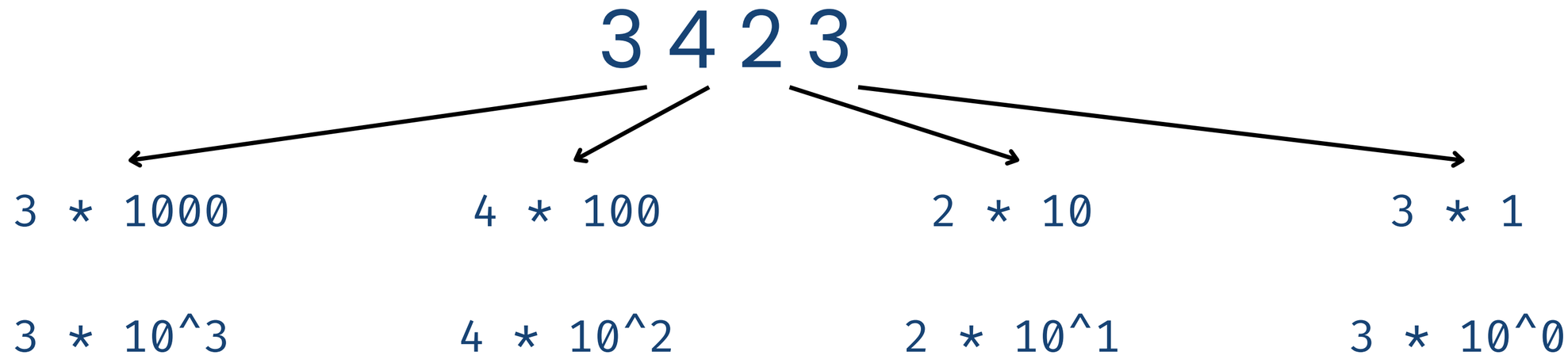
Let’s try for 135.
We have 13 sets of 10. but 10 sets of 10 is 100, so we now jump to a bigger number set. 1 set of 100, 3 sets of 10 and 5 single.

Each time we reach a set of ten, we name it something and make a new set.



11 = 10 + 1

Decimal Number



When the value of a digit depends on its position within the number, it is called a **positional number system**.

Decimal system is a positional number system, where each time we move left, the value is multiplied by **10 times** the previous one.

Exercise

Create a base-3 counting system and show the symbols for first 15 numbers.

0	10	20	100	110
1	11	21	101	111
2	12	22	102	112

Newton School of Technology

Create a base-5 counting system and show the symbols for first 15 numbers.

0	10	20
1	11	21
2	12	22
3	13	23
4	14	24

Newton School of Technology

Compare different numbers in base-10, base-3, base-5 and base-16 number systems.

Base 10	Base 3	Base 5	Base 16
32	1012	112	20
104	10212	404	68
11	102	21	B

For more: https://www.rapidtables.com/convert/number/base-converter.html

Binary Number System

Two Symbols

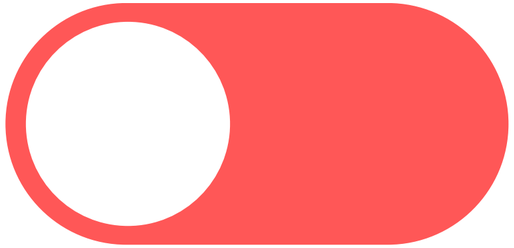
As the most basic electronic devices can have 2 states, on or off, they can be represented very easily using binary numbers 0 and 1.

Hence, the binary numbers play a critical role in design of digital electronic systems.

1

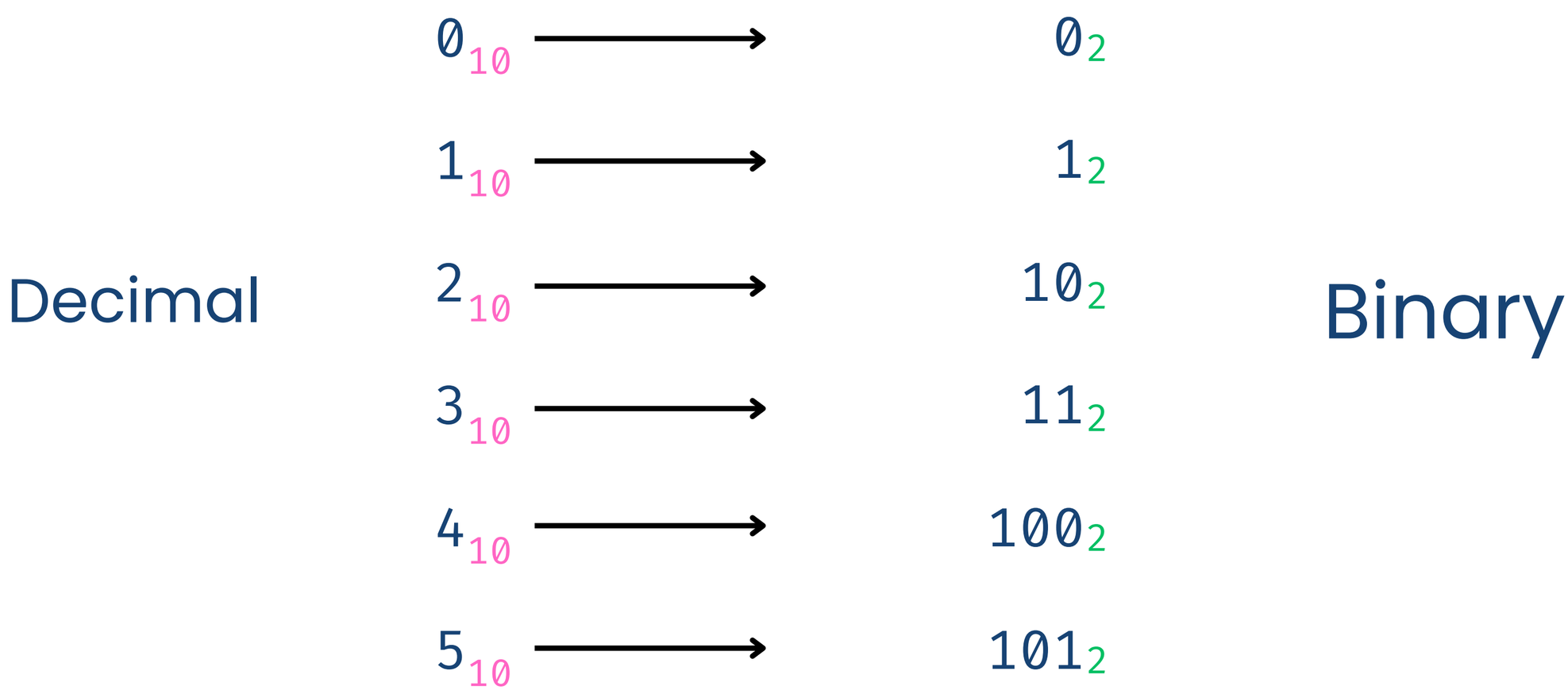


0

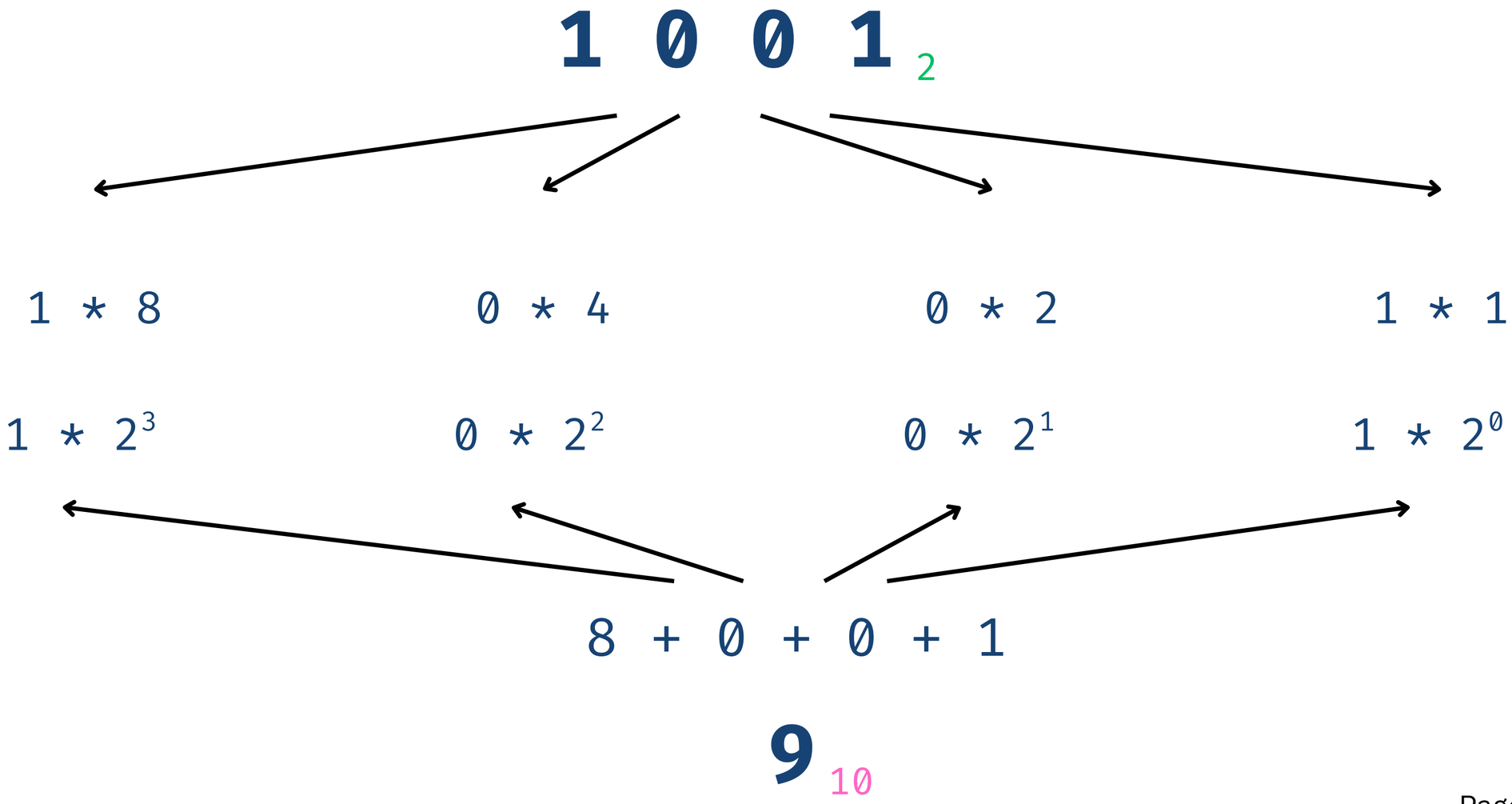


How We Count in Binary

Newton School of Technology



Binary to Decimal

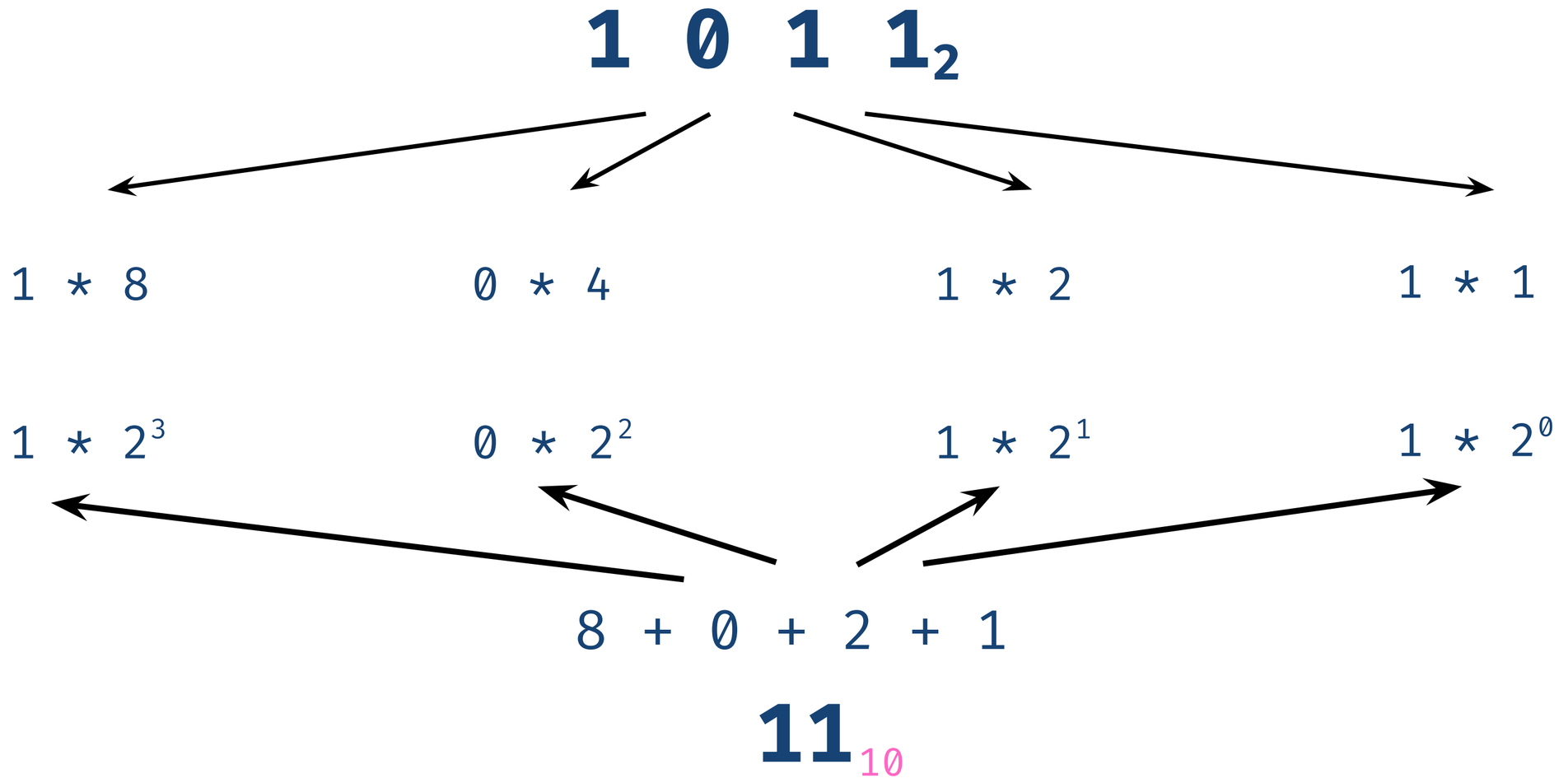


Newton School of Technology

What Number Is This?

1011_2

Binary to Decimal



Decimal to Binary

Step 1: Divide the given number 13 repeatediy by 2 until you get '0' as the quotient

$13 \div 2 = 6$ (Remainder 1)

$6 \div 2 = 3$ (Remainder 0)

$3 \div 2 = 1$ (Remainder 1)

$1 \div 2 = 0$ (Remainder 1)

Step 2: Write the remainders in the reverse order

1 1 0 1

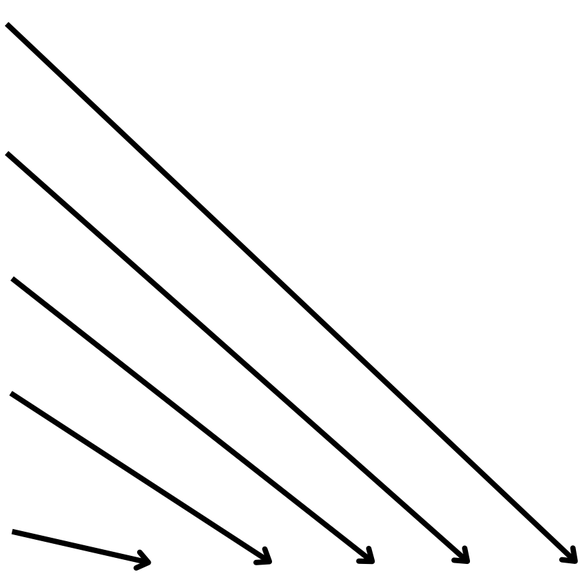
$\therefore 13_{10} = 1101_2$

(Decimal) (Binary)

Exercise

25$_{10}$

	Q	R
25/2 =	12 ,	1
12/2 =	6 ,	0
6/2 =	3 ,	0
3/2 =	1 ,	1
1/2 =	0 ,	1



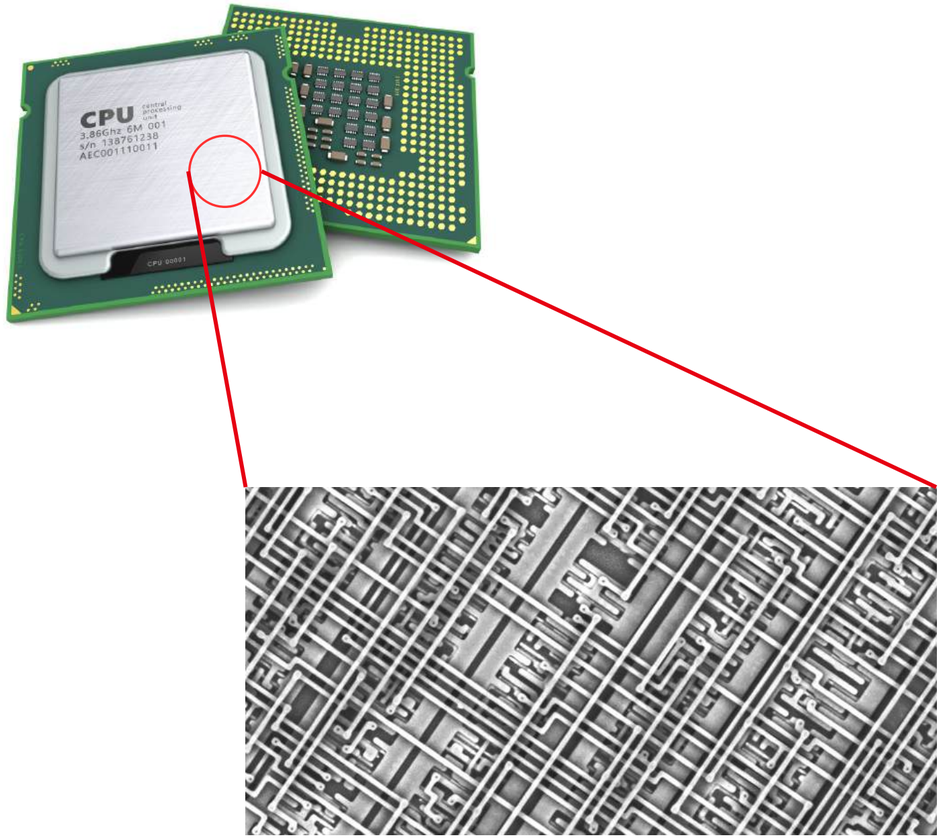
1 1 0 0 1$_2$

Newton School of Technology

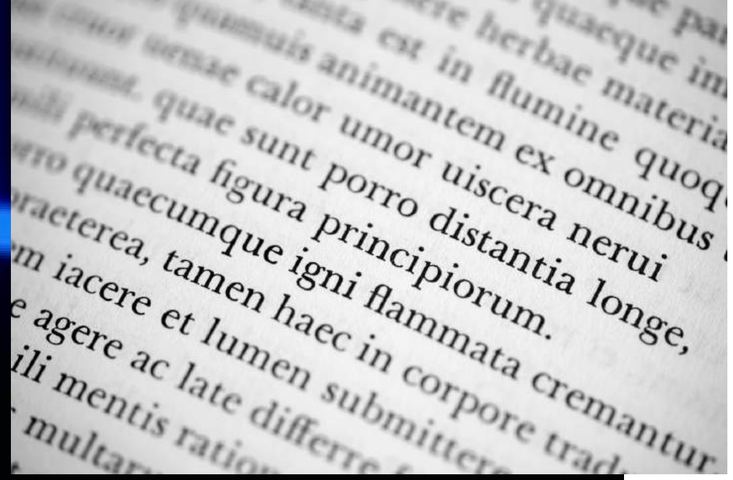
How Does a Computer Store a 0 or 1?

Computers & Numbers





Computers are **electronic** devices that process information by turning billions of **transistors** on and off, representing **binary data** as 1s and 0s.







Computers use numbers to represent instructions, text, images, and all other forms of data.

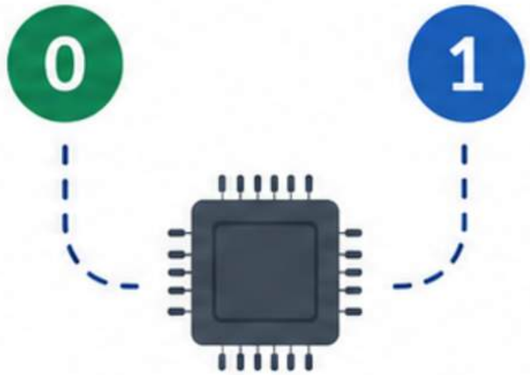
There are different types of numbers such as decimal, binary, hexadecimal.

Meet the Bit

- A bit is the smallest unit of digital information.

The word bit means:

Binary Digit





Computers combine many bits to represent larger values and more complex data.

A bit can store one of two values:

0 or **1**

How Much Can Bits Represent?

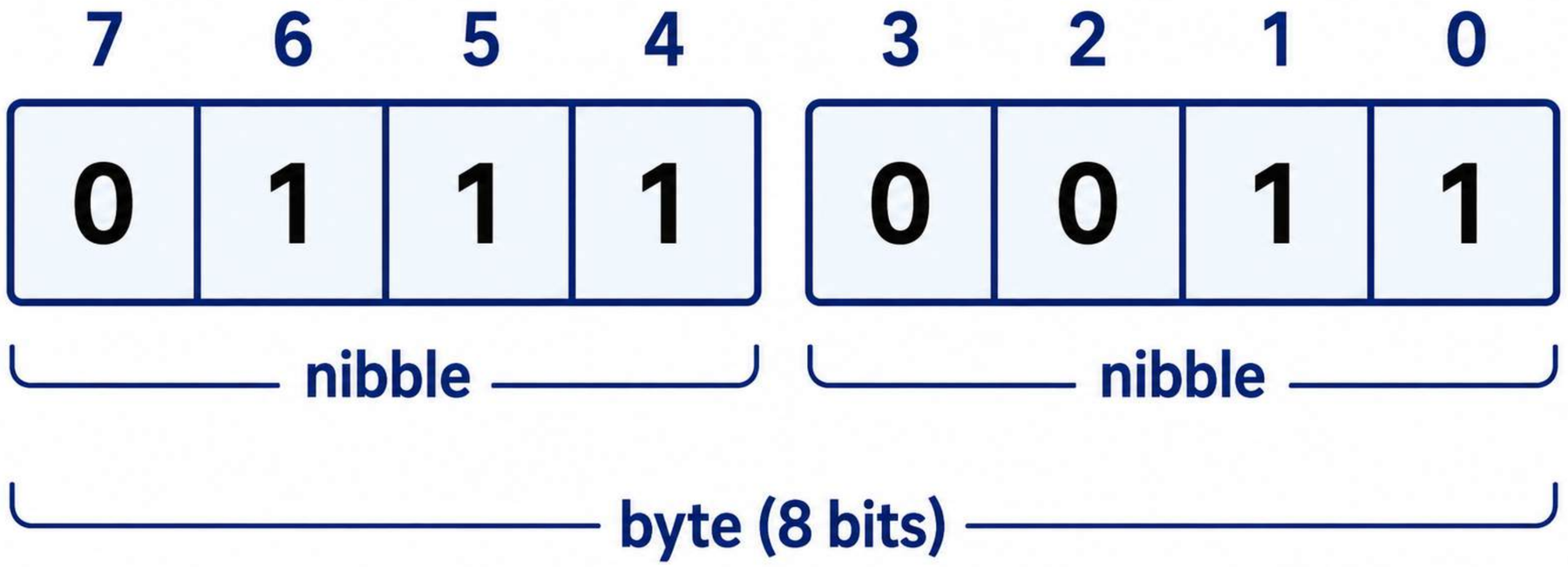
Newton School of Technology

- A single bit can have only two values: 0 or 1.
- When we combine more bits, the number of possible combinations increases. With n bits, we can represent 2^n different values.

Number of Bits	Possible Values (2^n)	Unsigned Range
1 bit	$2^1 = 2$	0 to 1
2 bits	$2^2 = 4$	0 to 3
3 bits	$2^3 = 8$	0 to 7
4 bits	$2^4 = 16$	0 to 15
...	...	...
8 bits	$2^8 = 256$	0 to 255

How Bits Are Grouped

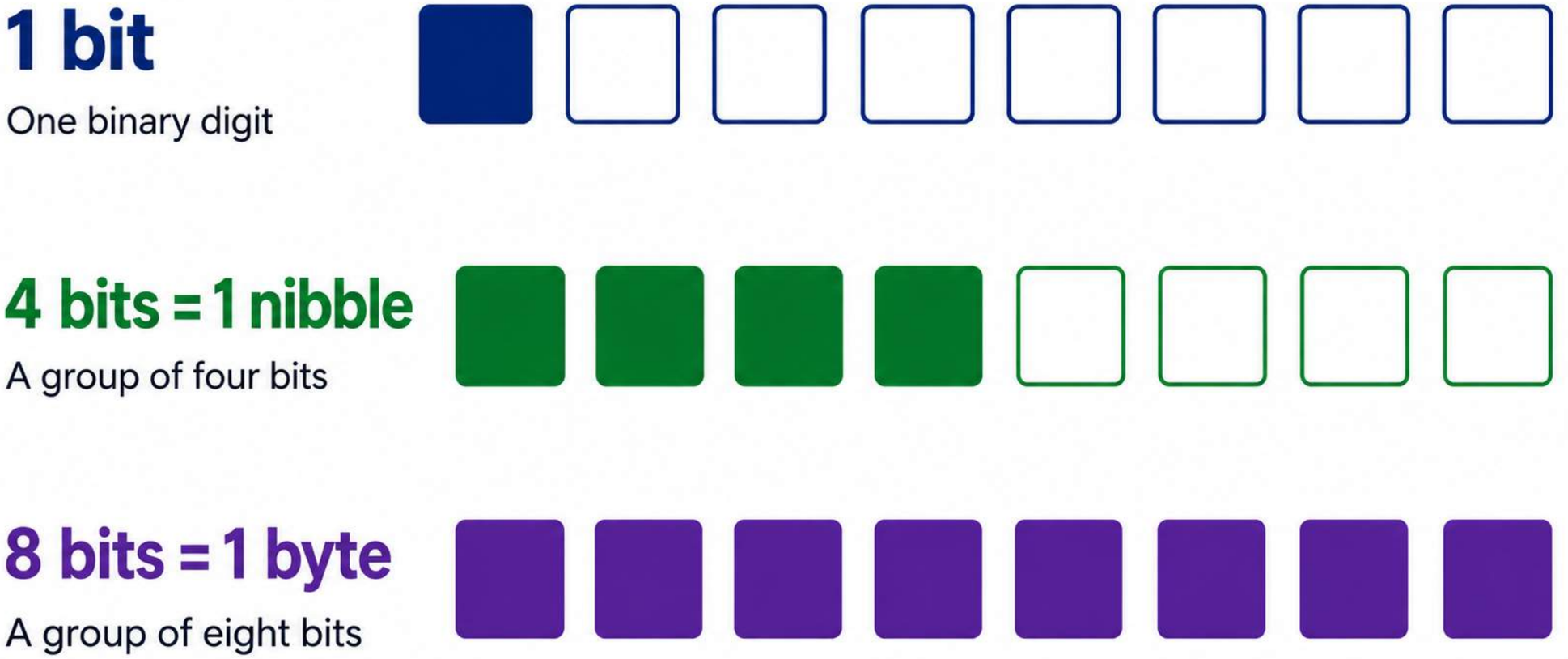
- Computers combine individual bits into standard units.
- These groups make data easier to store and process.



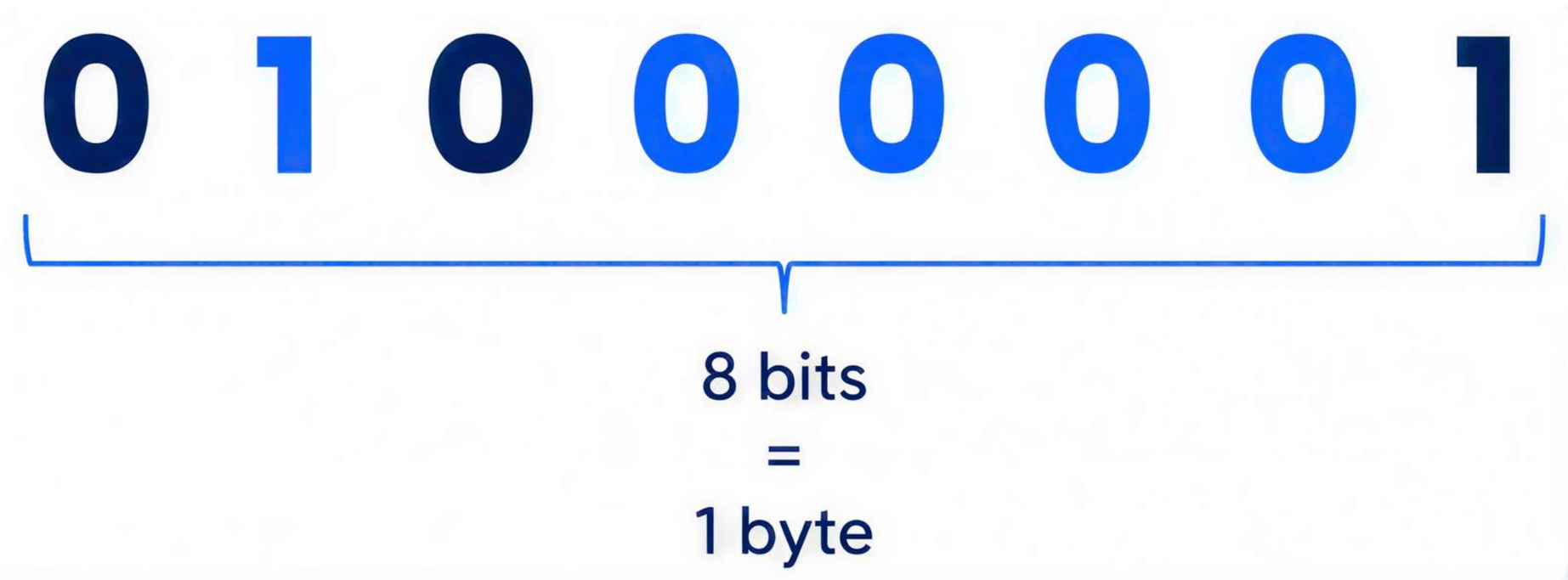
1 byte = 2 nibbles = 8 bits

Common Bit Units

- A bit is the smallest unit of digital information.



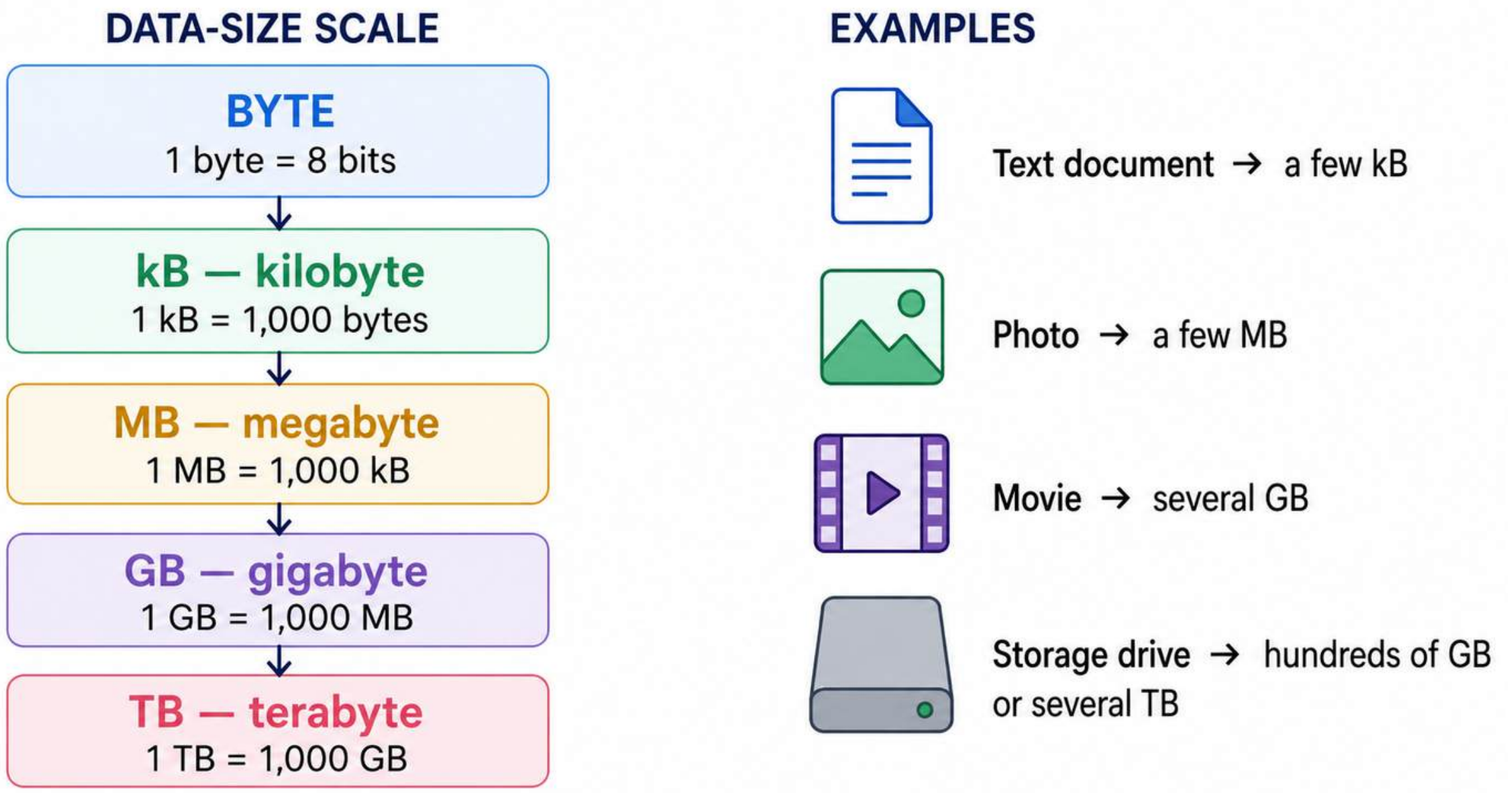
What Can One Byte Represent?



By itself, this is only a pattern of 0s and 1s.

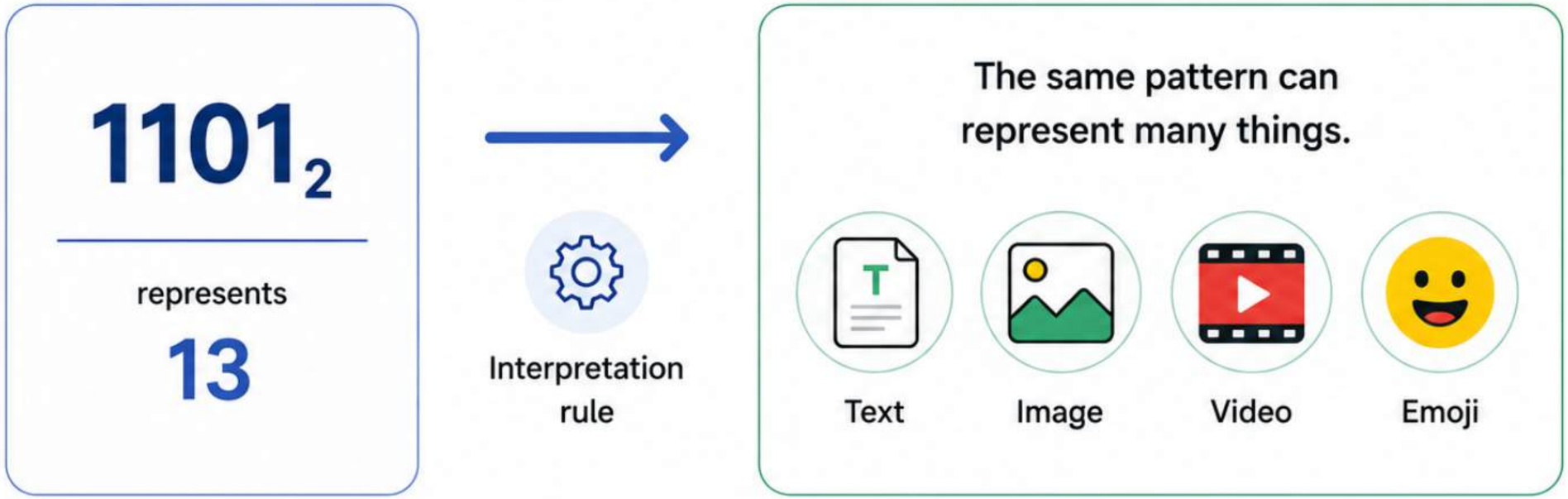
From Bytes to Larger Storage Units

- A byte contains 8 bits.
- Storage capacity is commonly measured using decimal units.



From Numbers to Data

- How can the same 0s and 1s represent text, images, and videos?



As a number

As other data

How Can Binary Represent Text?

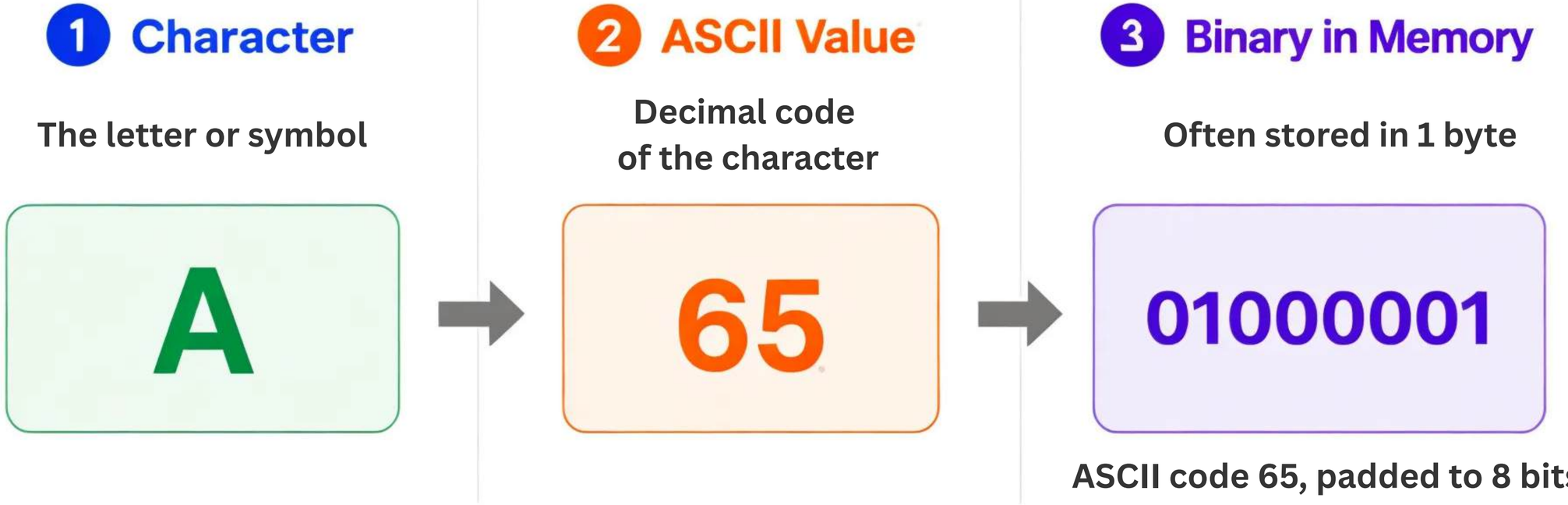
- To store text, computers use an agreed mapping:
  - Each character is assigned a number.



ASCII: Giving Numbers to Characters

Newton School of Technology

- ASCII (American Standard Code for Information Interchange) assigns a unique number to each character.
- Standard ASCII defines 128 characters using 7 bits.



Newton School of Technology

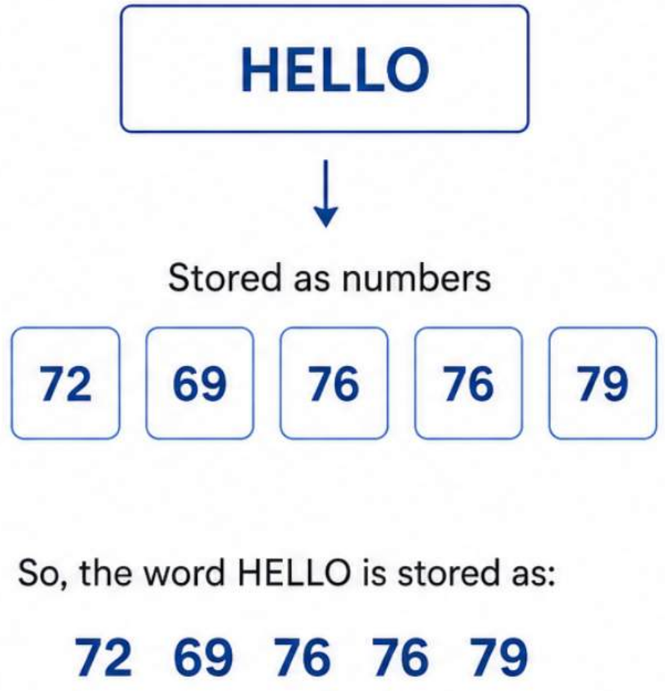
Can You Convert "HELLO" into ASCII?

Encoding a Word with ASCII

- ASCII assigns a number to each character.
- To store a word, the computer stores the ASCII number of every character in sequence.

Example:

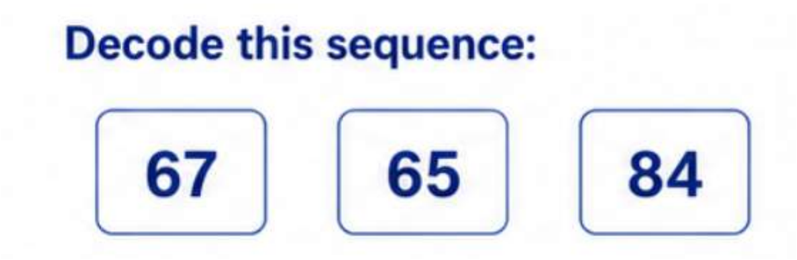
Character	ASCII Code
H	72
E	69
L	76
L	76
O	79



Decode the Hidden Word

Newton School of Technology

- If computers can convert characters into numbers, they can also convert numbers back into characters using the same ASCII mapping.



Decode the Hidden Word

Newton School of Technology

- If computers can convert characters into numbers, they can also convert numbers back into characters using the same ASCII mapping.

Decode this sequence:

67 65 84

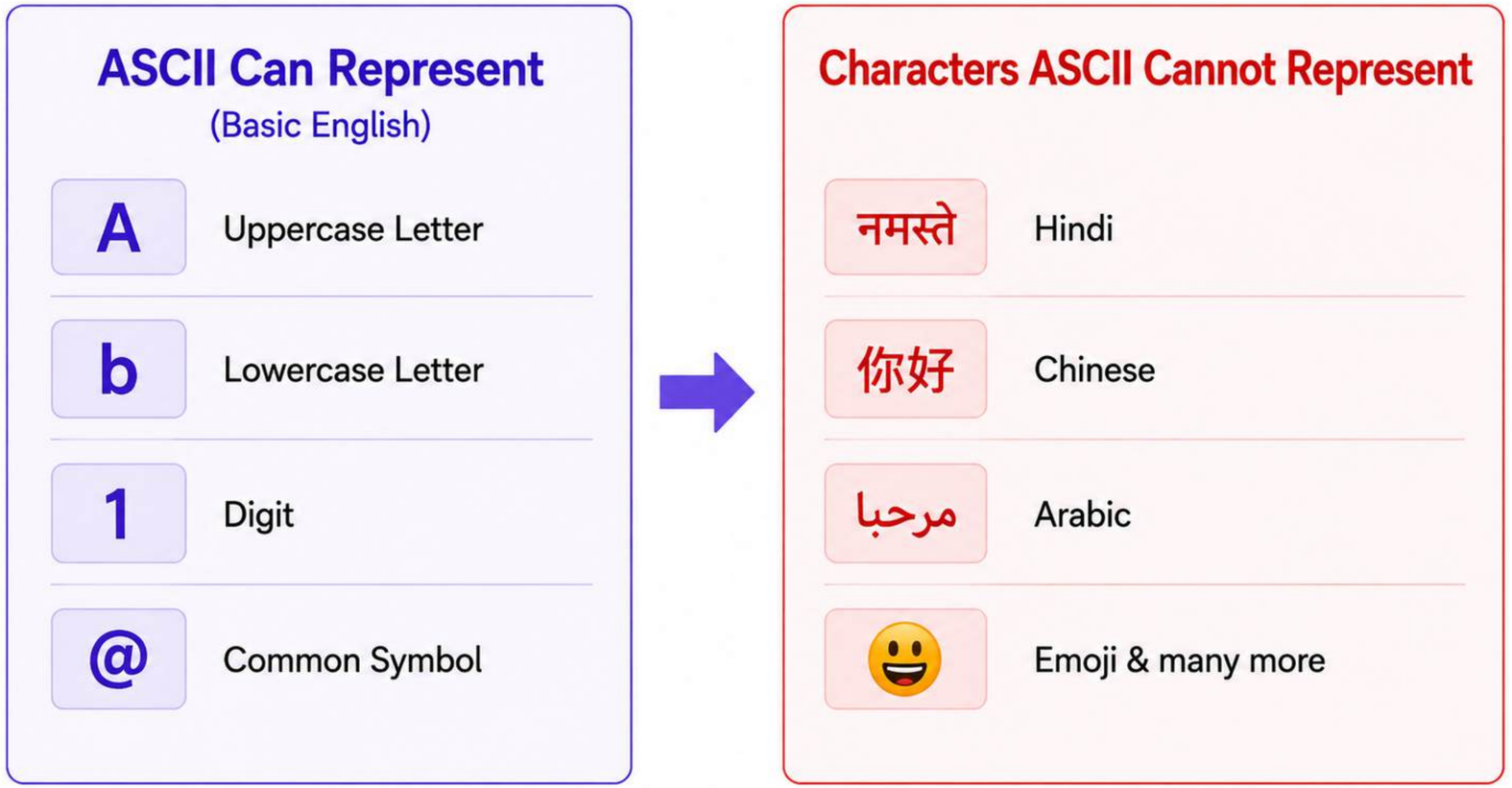
Using the ASCII table:

ASCII Code	Character
67	C
65	A
84	T

→ **CAT**

ASCII Is Not Enough

- ASCII could represent English letters, numbers, and symbols.
- But computers soon needed to store text from every language in the world.

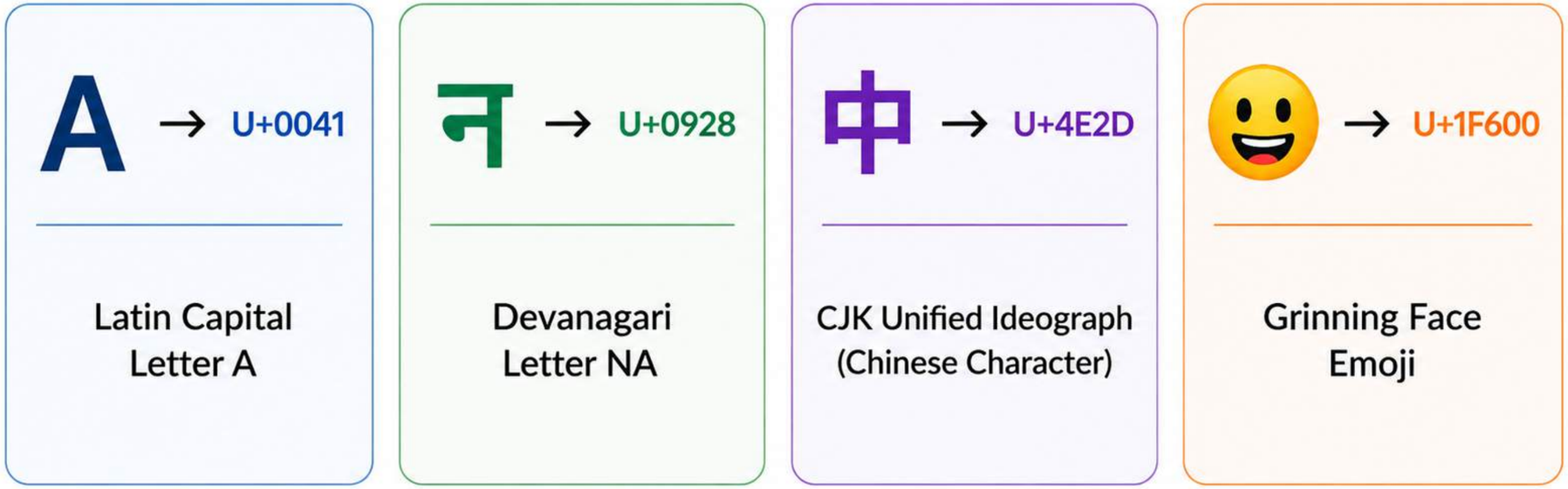


Unicode Gives Every Character a Code

Newton School of Technology

- Unicode assigns code points to encoded characters and text symbols.
- It supports characters from **different languages and symbols** used all over the world.

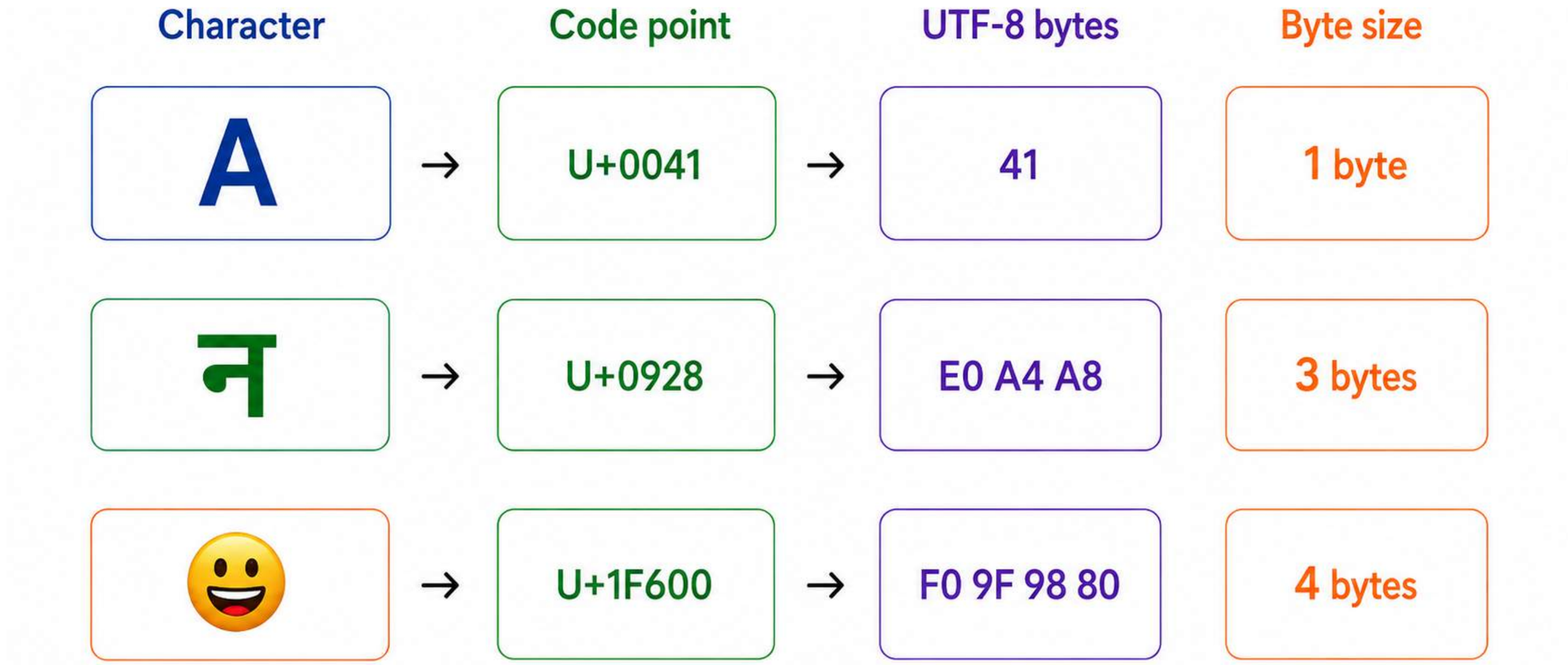
Example: Characters from different languages all receive unique code points.



UTF-8 Converts Unicode into Bytes

Newton School of Technology

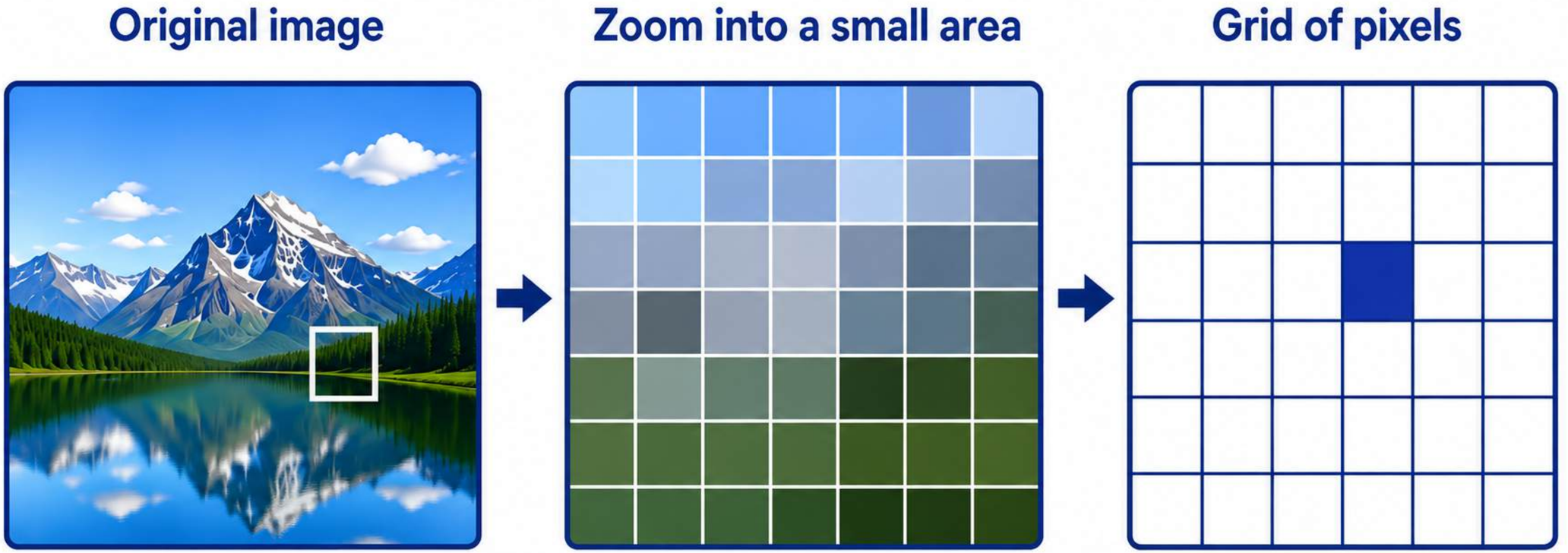
- Unicode identifies the character.
- UTF-8 converts it into bytes for storage and transmission.



How Can Binary Represent an Image?

Newton School of Technology

- Digital photographs are represented as a grid of tiny squares called **pixels**.
- Thousands or millions of **pixels** together form the complete **image**.

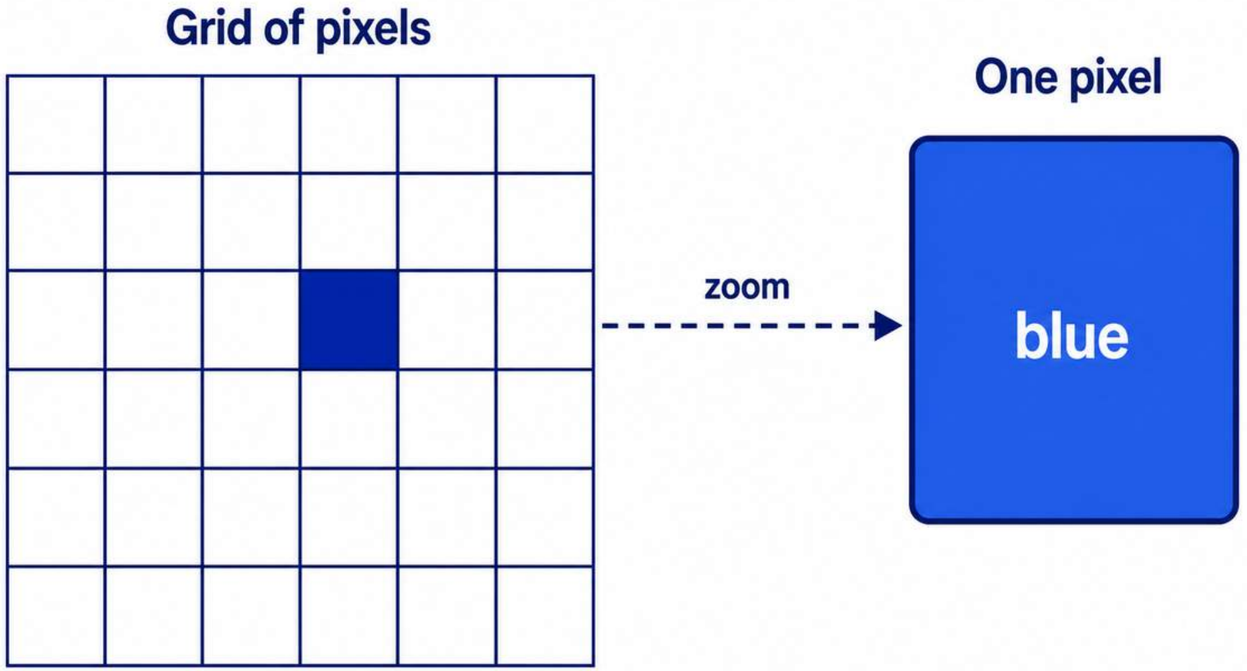


A small area is a grid of pixels

How Can Binary Represent an Image?

Newton School of Technology

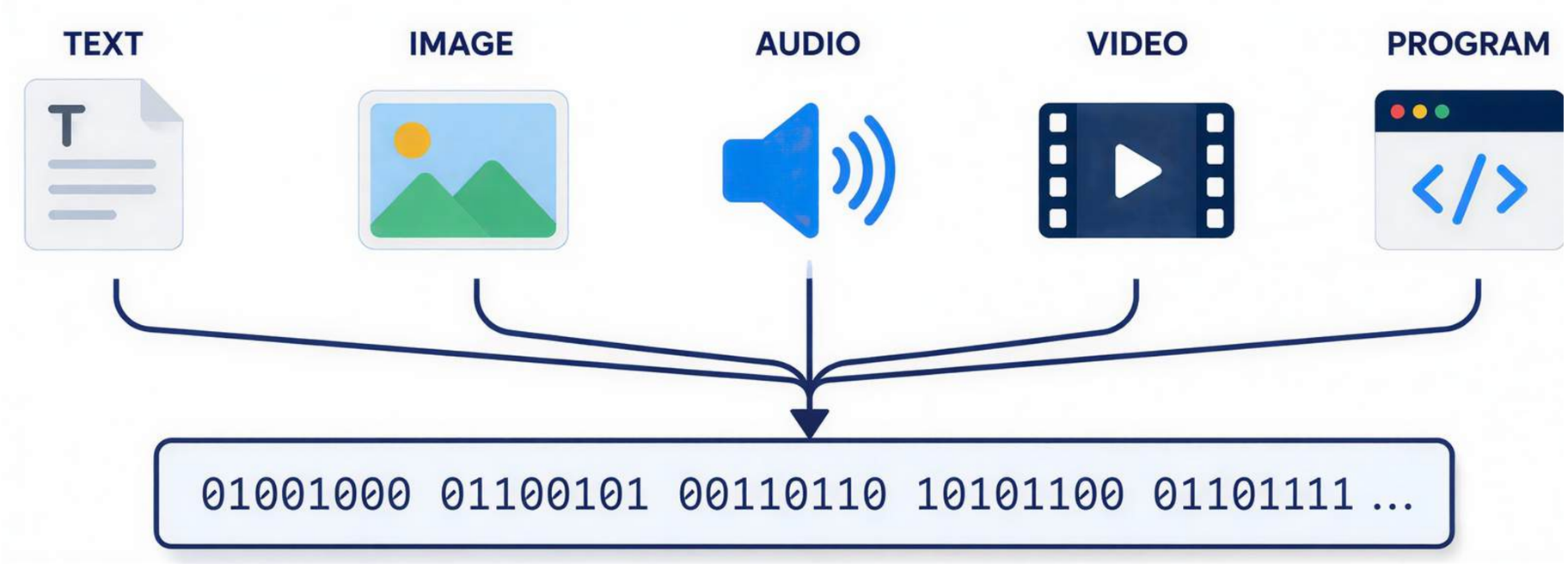
- Each **pixel** stores numeric values that describe its **colour**.



Each tiny square (pixel) has a colour represented using numbers (binary).

Everything Becomes Bytes

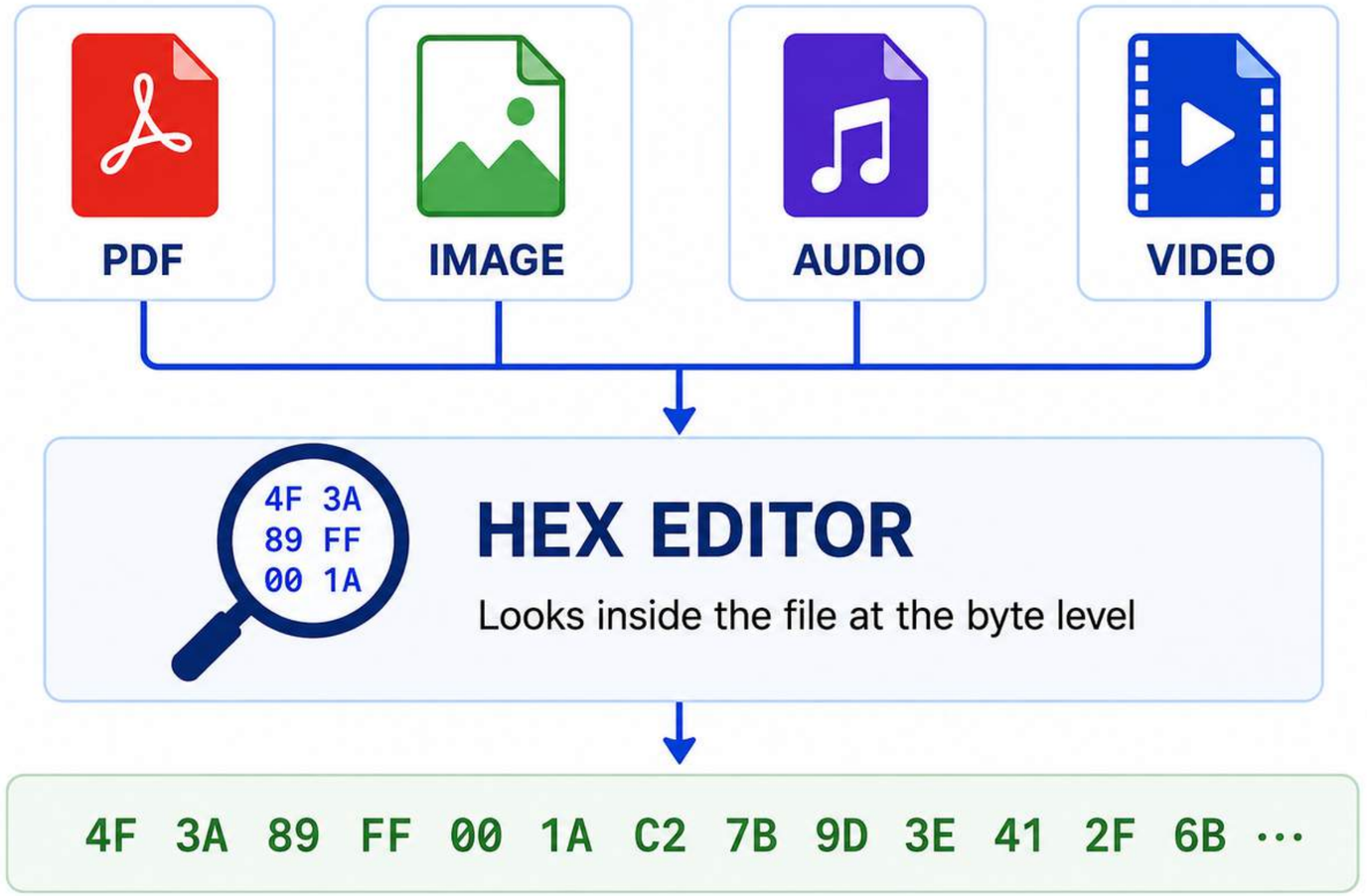
- Text, images, audio, video, and programs may look different to us.
- Inside a computer, they are all stored as **sequences of bytes.**



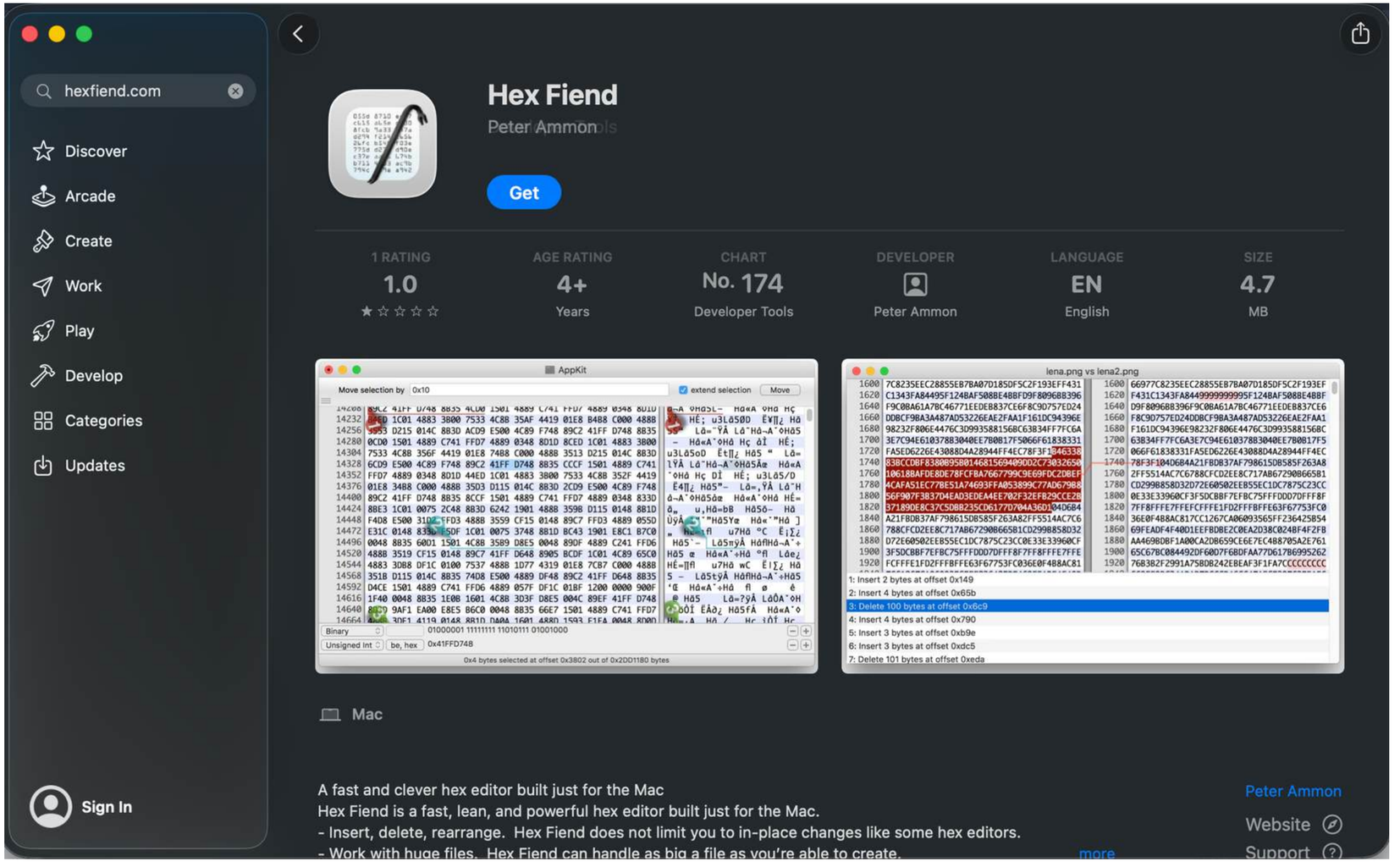
Same building blocks. **Different meanings**.

Can We Actually See These Bytes?

- A **hex editor** lets us inspect the bytes stored inside a file.

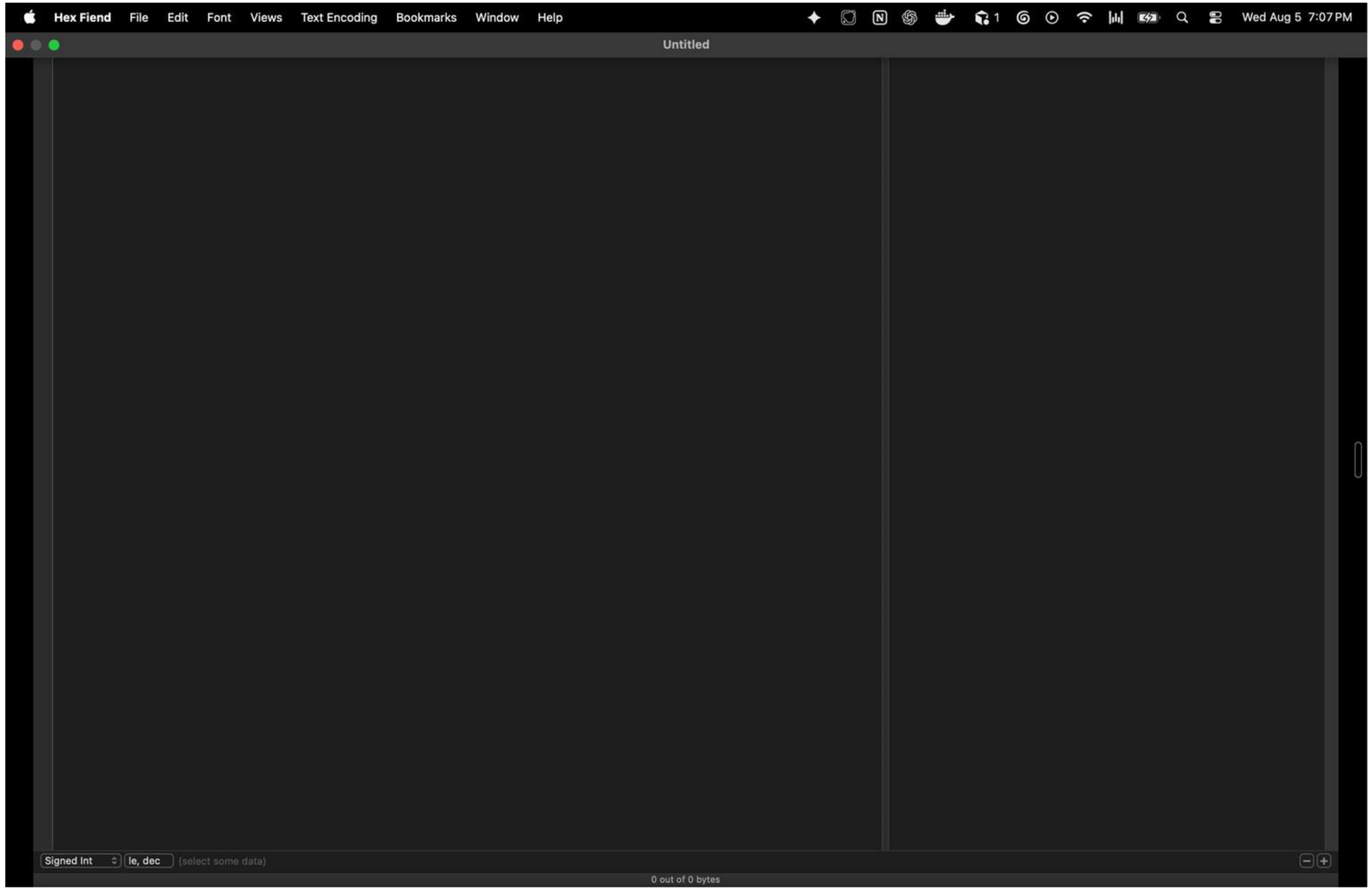


Download Hex Fiend

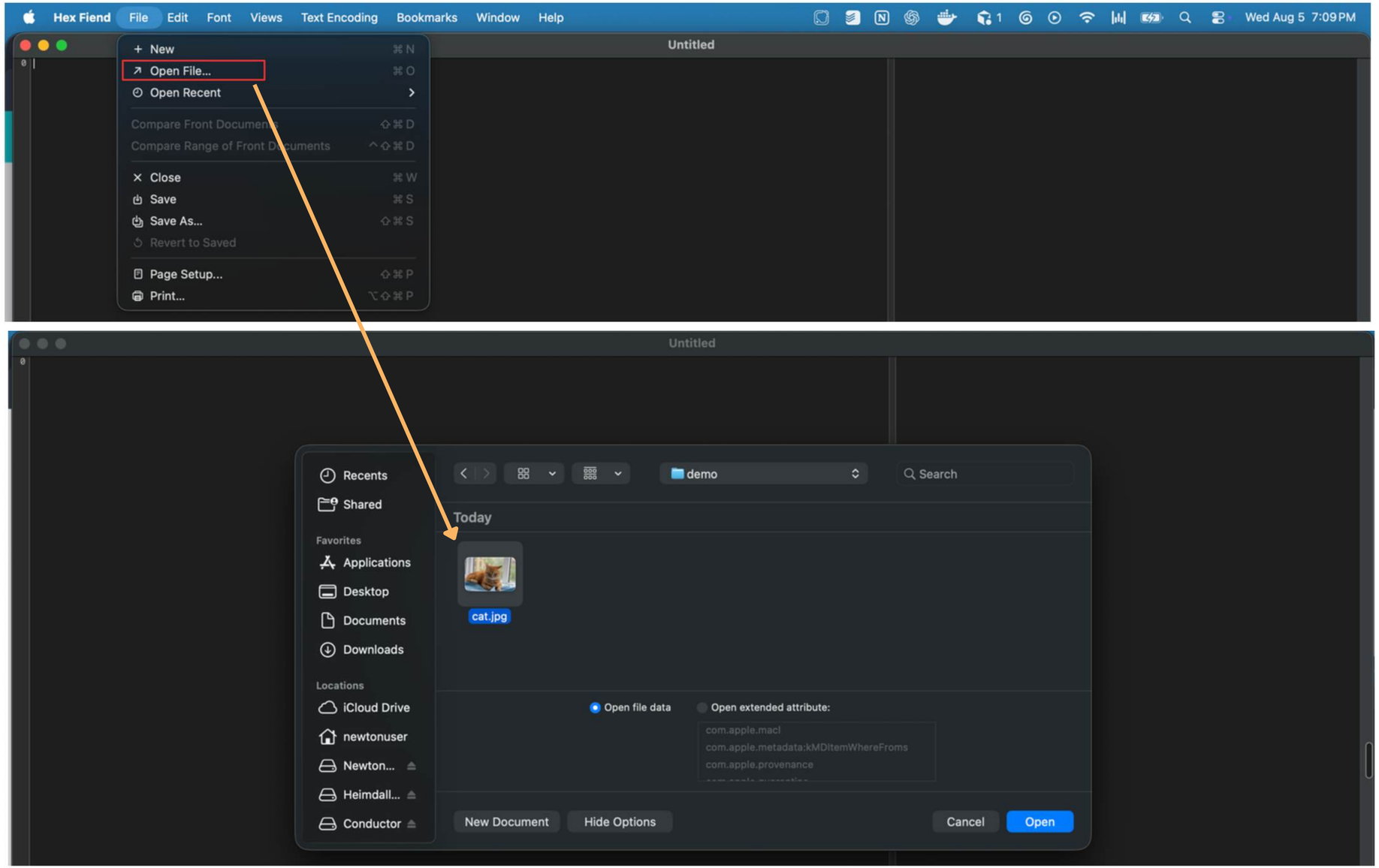


Launch the app

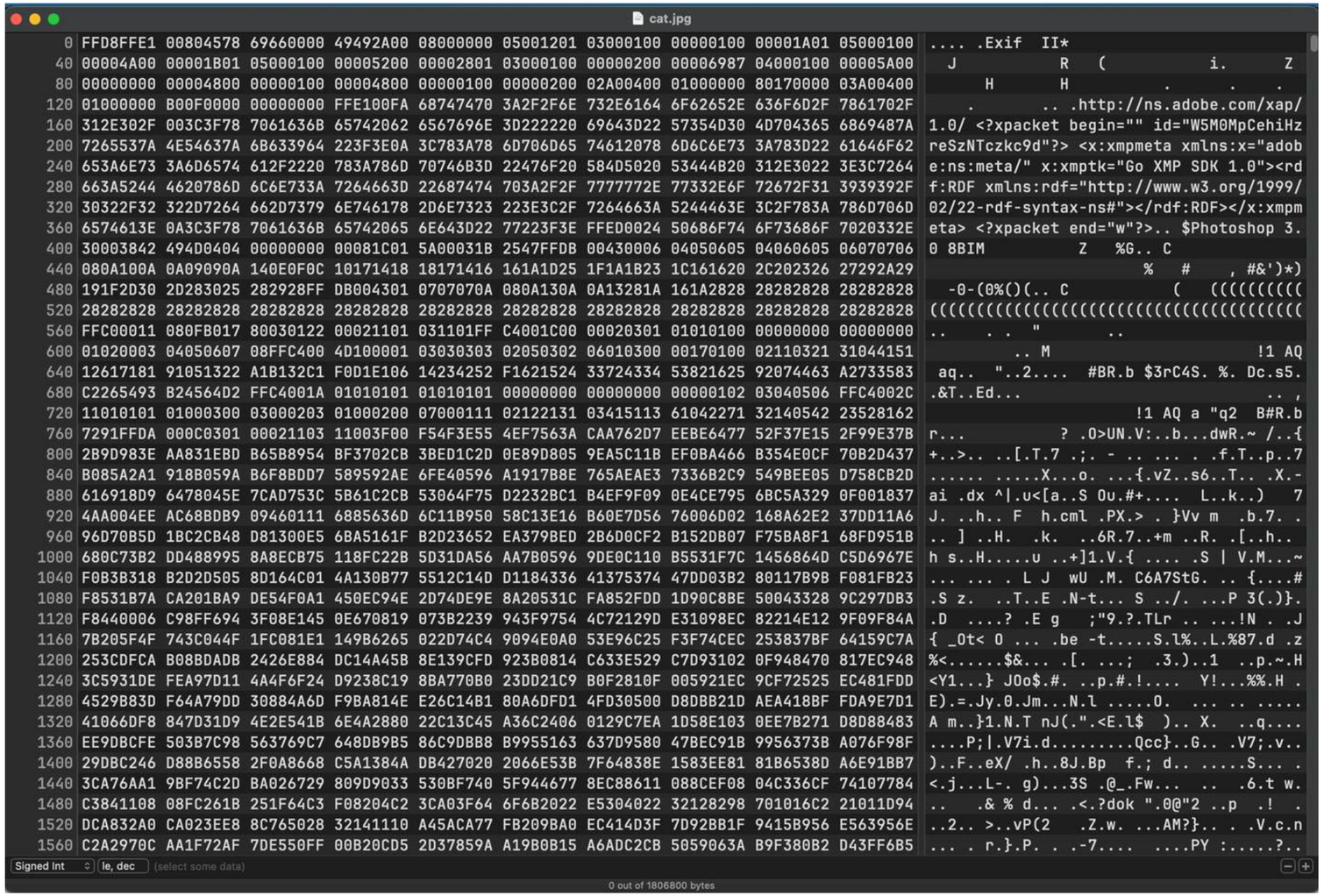
Newton School of Technology



Open the Image



Meet the Hex Editor



Wait... This Isn't Binary!

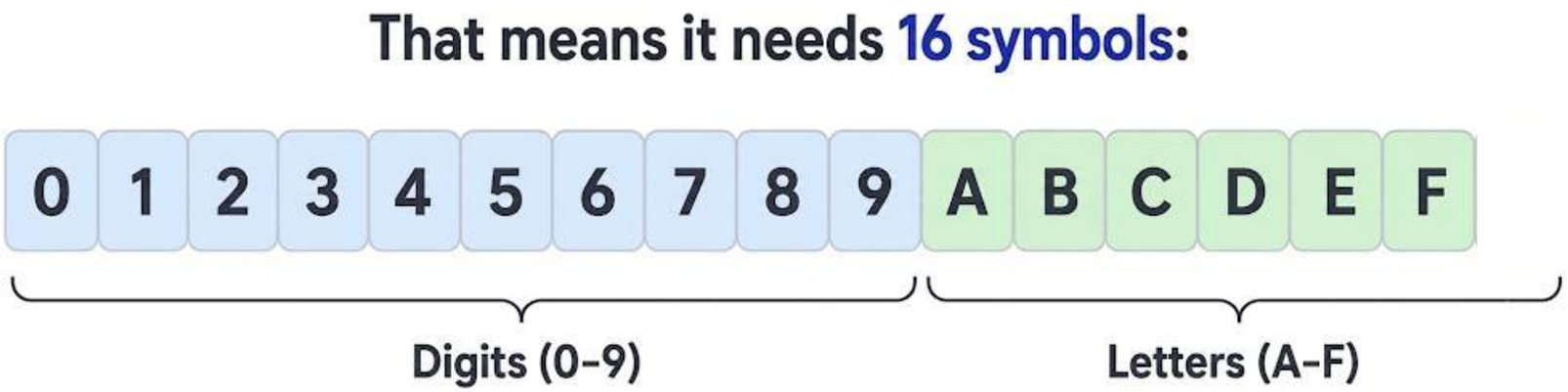
- Computers store data in binary, but hex editors display the same bytes in hexadecimal.



The data is still binary. Only its display has changed.

Hexadecimal Number System

- Hexadecimal is a **base-16** number system.



A = 10 **B** = 11 **C** = 12

D = 13 **E** = 14 **F** = 15

Four Binary Bits Make One Hex Digit

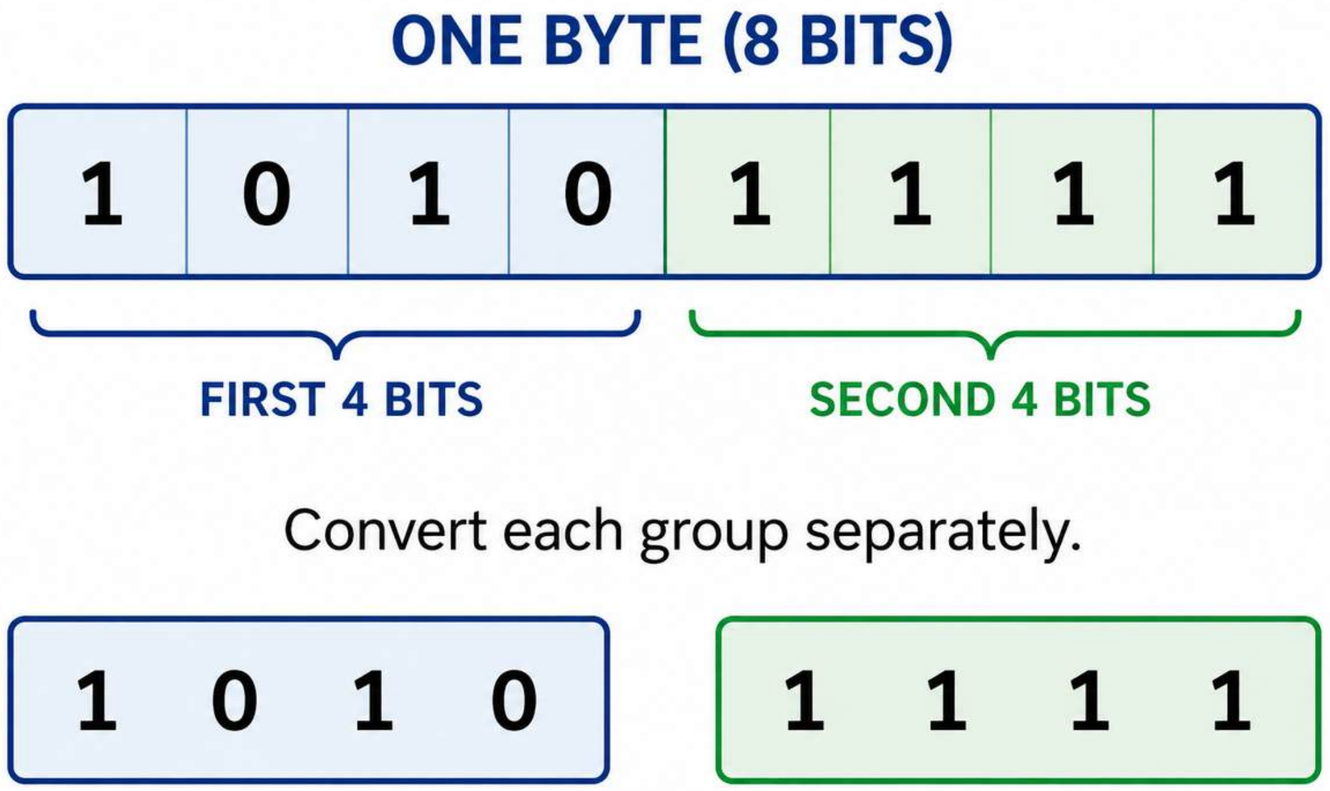
Newton School of Technology

- Every group of **4 binary bits** can be represented by **1 hexadecimal digit.**

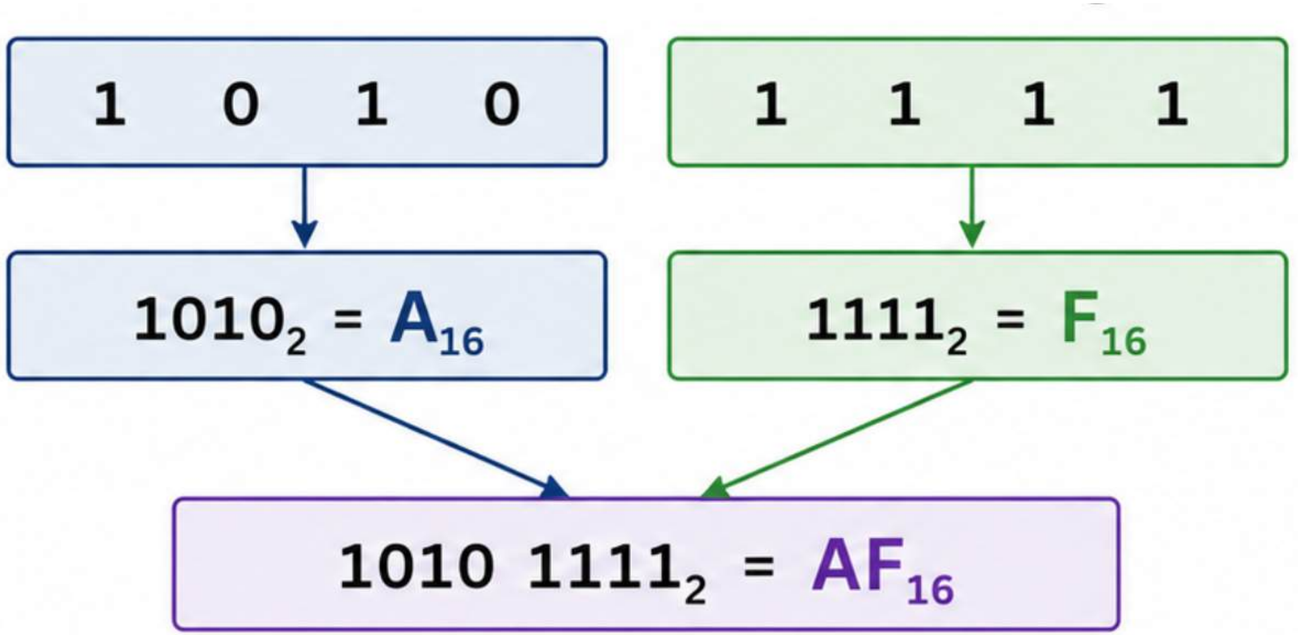


One Byte Becomes Two Hex Digits

- A byte contains **8 bits**. Split the byte into two groups of four.



One Byte Becomes Two Hex Digits



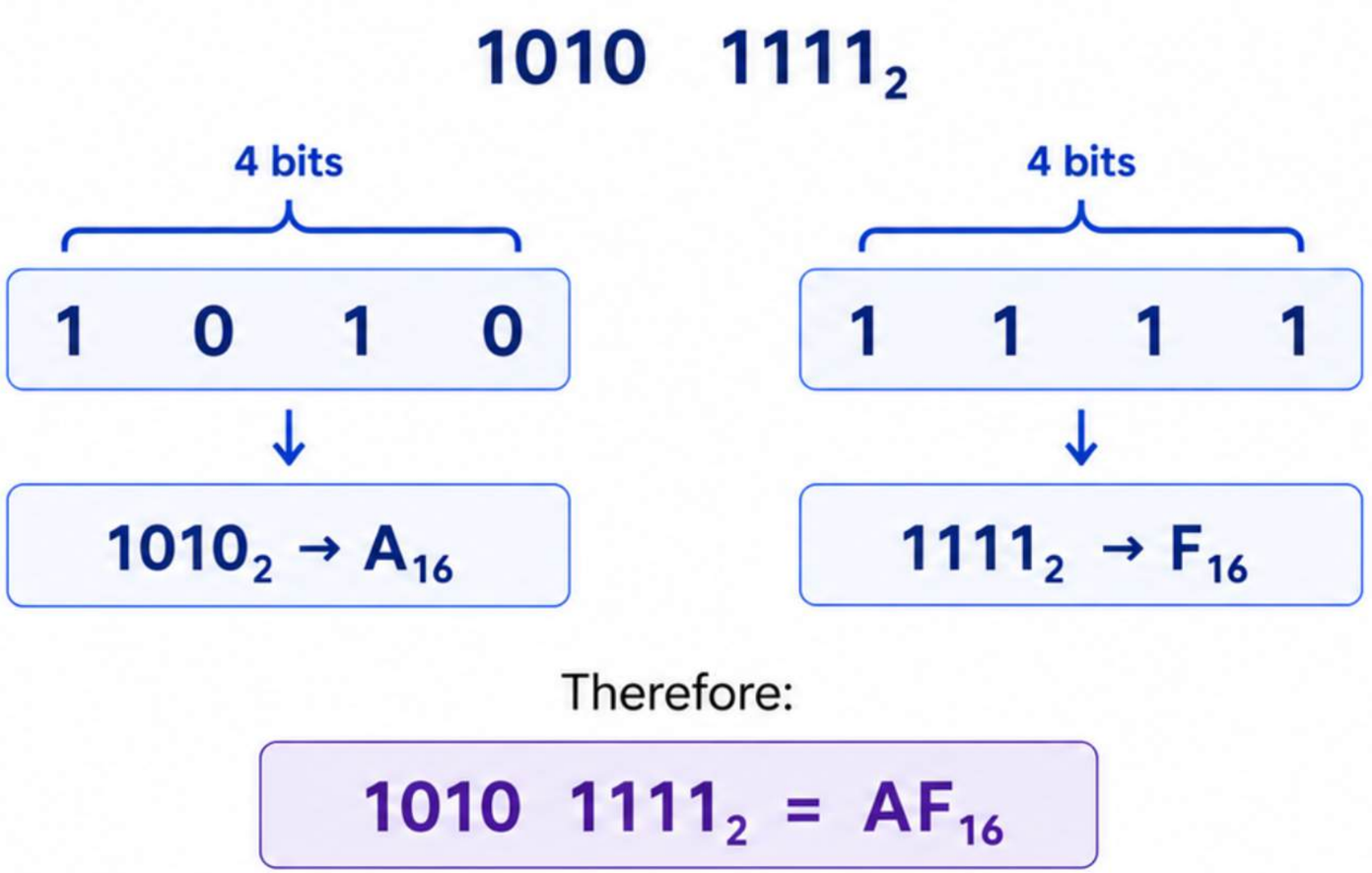
8 bits = 1 byte = 2 hex digits

Bytes are commonly displayed as pairs:

4A	FF	89	2C

Binary to Hexadecimal

- Starting from the right, split the binary number into groups of **4 bits.**
- Convert each group into **one hexadecimal digit.**



Where Do We See Hexadecimal?

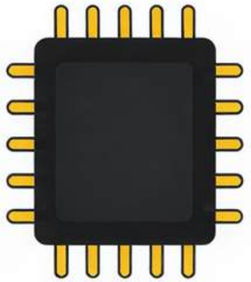
- Hexadecimal is commonly used when binary data needs to be shown in a shorter and more readable form.



Color codes

`# FF 5733`

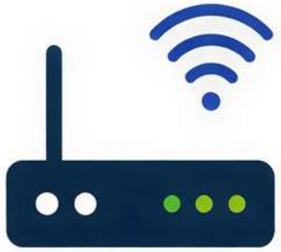
(Red–Orange)



Memory addresses

`0x7FFFE`

(Location of data in memory)



MAC addresses

`00:1A:2B:3C:4D:5E`

(Network device identity)



File bytes

`89 50 4E 47 0D 0A 1A 0A`

(Beginning of a PNG file)

Further Reading

Newton School of Technology

- **tutorialspoint**

  [https://www.tutorialspoint.com/basics_of_computers/basics_of_computers_number_system.htm]
- **Wikipedia - Numeral system**

  [https://en.wikipedia.org/wiki/Numeral_system]
- **Base Converter**

  [https://www.rapidtables.com/convert/number/base-converter.html]
- **ASCII Table**
- **[**https://simple.wikipedia.org/wiki/ASCII**]**

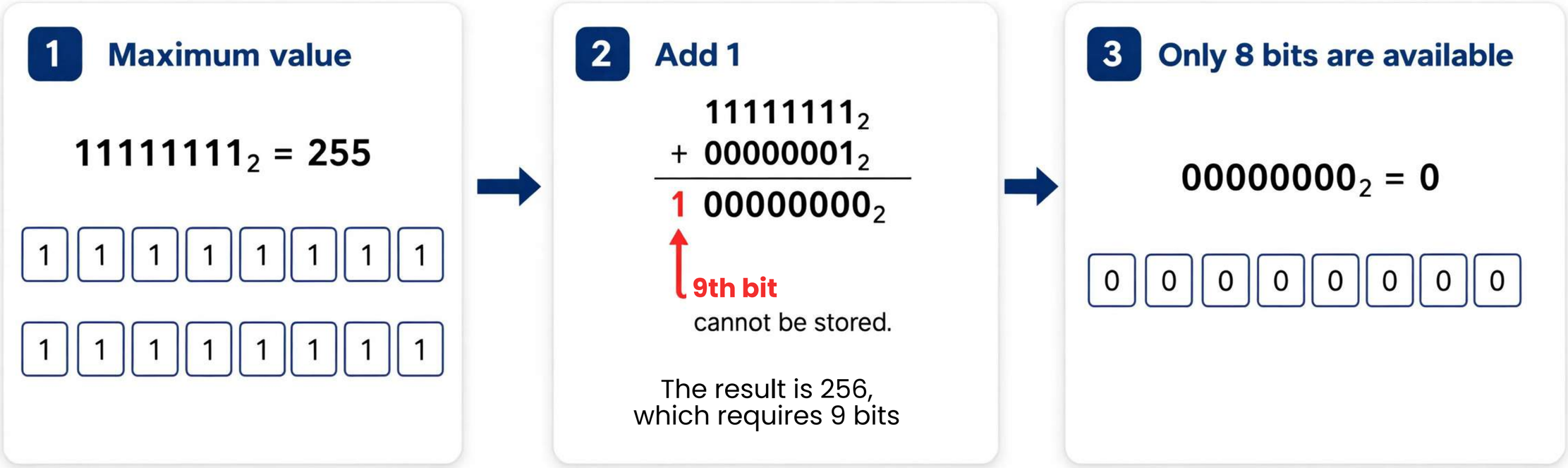
Thanks for watching!

Newton School of Technology

Deep Dive

Binary Overflow

- An unsigned 8-bit value can represent numbers from 0 to 255.

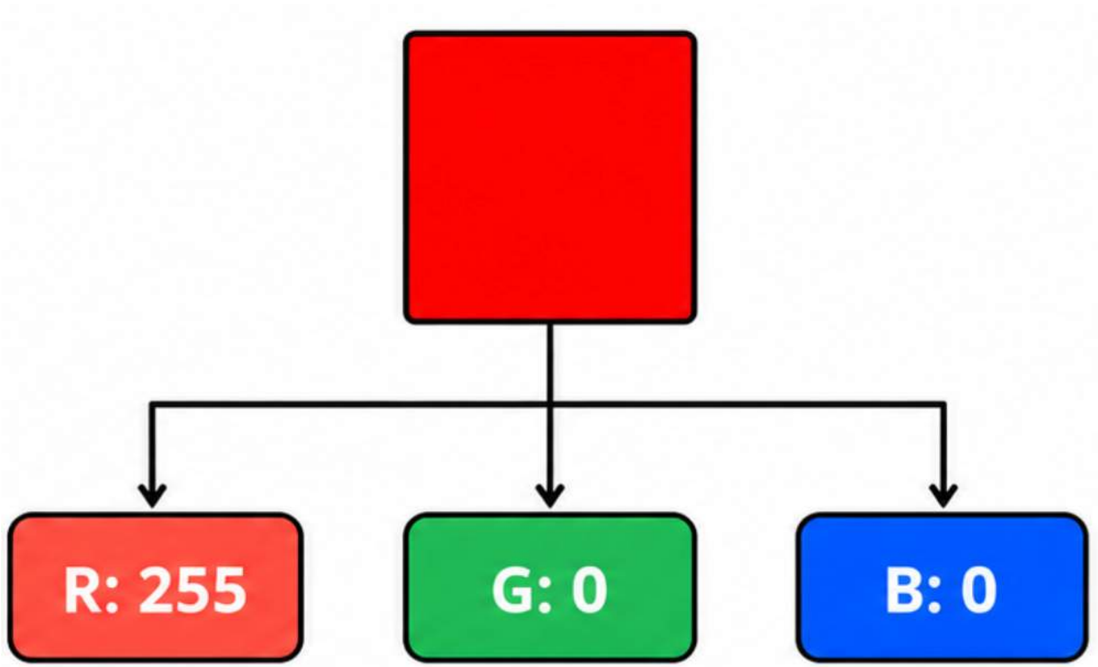


The value wraps from 255 to 0. This is called **overflow**.

RGB: A Pixel Stores Three Values

Newton School of Technology

- Every colour on your screen is created using three values: **Red**, **Green**, and **Blue**.
- In common 8-bit RGB, each channel ranges from 0 to 255.
- Each pixel stores three numbers:
  - R → **Red: 0-255**
  - G → **Green: 0-255**
  - B → **Blue: 0-255**



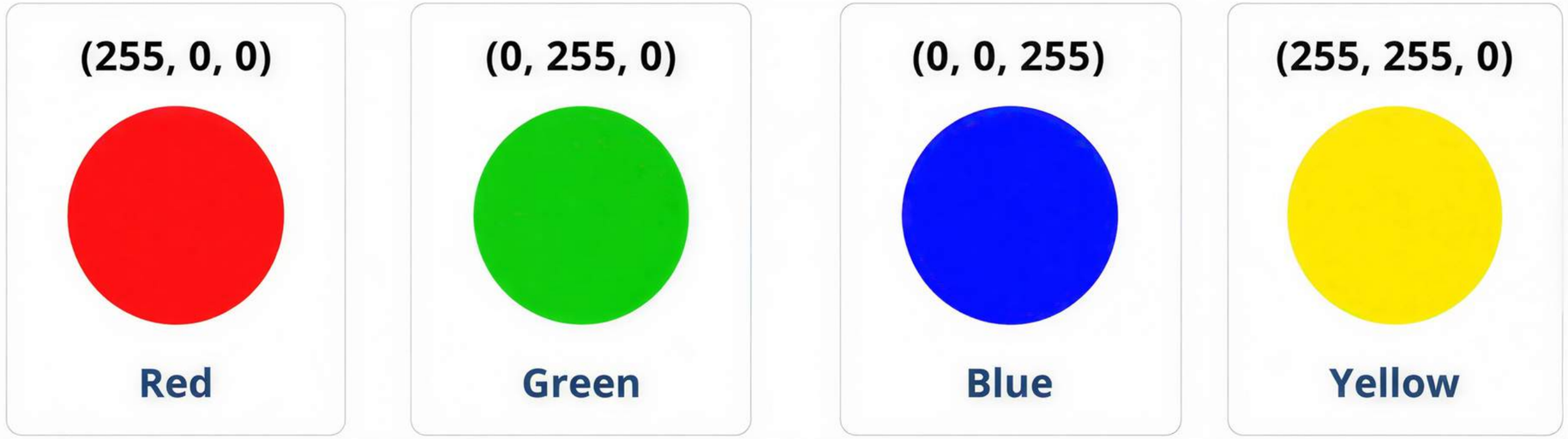
Example

RGB (255, 0, 0) = Red

Combining RGB Values Creates Colours

Newton School of Technology

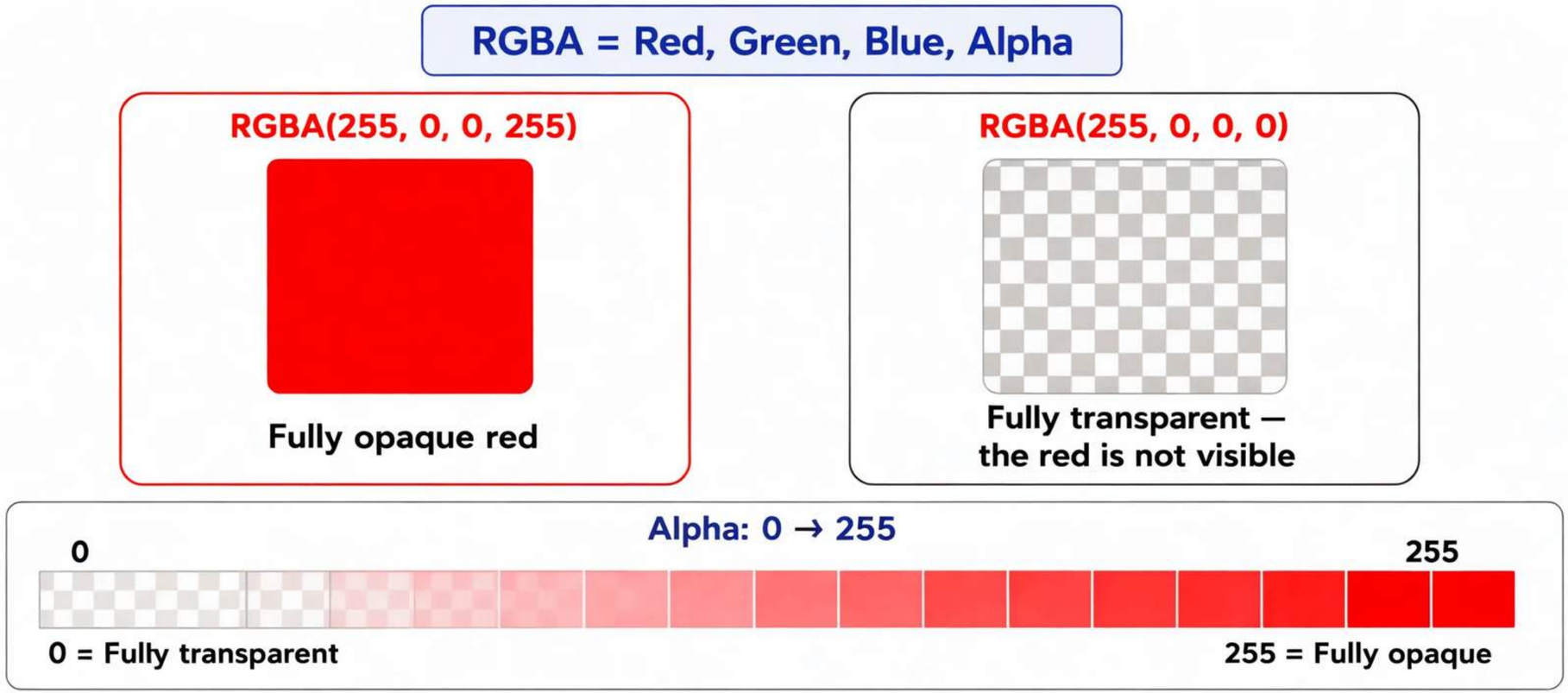
- Changing the three RGB values changes the colour.



Combining red, green, and blue
can create millions of colours.

What Is RGBA?

- RGB stores the colour of a pixel using three channels: Red, Green and Blue.
- In an 8-bit-per-channel image, each RGB value commonly ranges from 0 to 255.
- RGBA adds a fourth value called Alpha, which controls transparency.

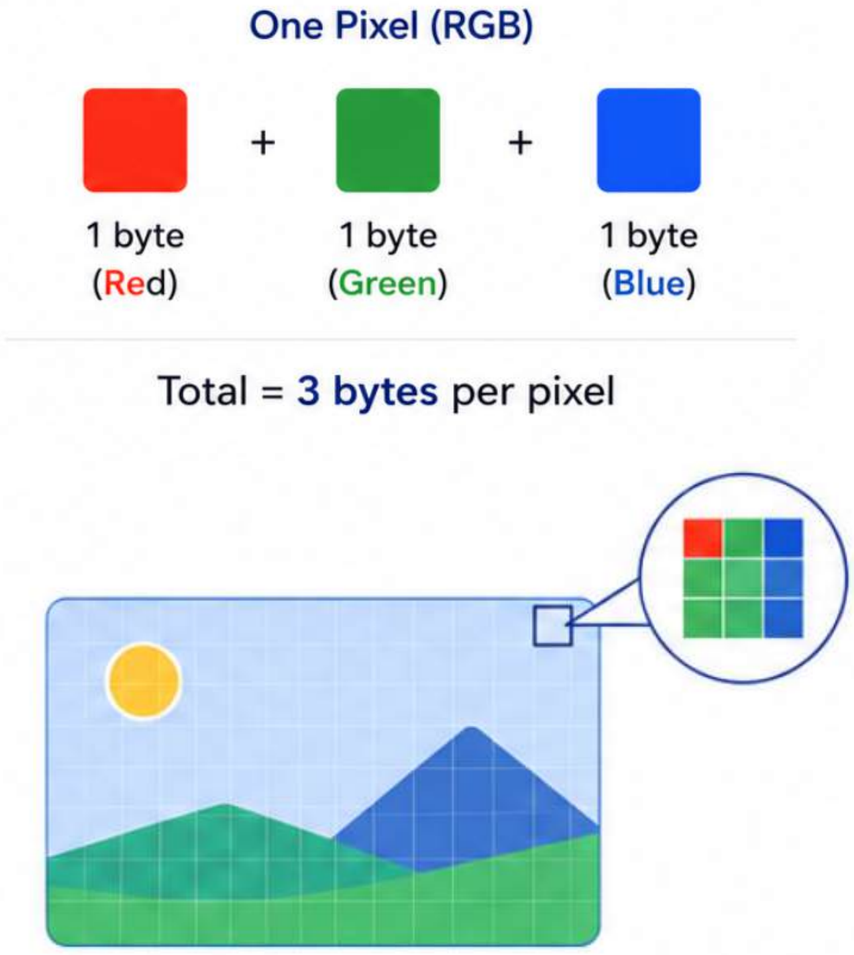


Uncompressed RGB Image Size

- Digital images are made of pixels.
- For a standard 8-bit RGB image, each pixel uses 3 bytes one byte each for **R**, **G** and **B**.
- Uncompressed RGB pixel data = width × height × 3 bytes

Example Calculation

Image size	=	1920 × 1080 pixels
Bytes per pixel	=	3
Total size	=	1920 × 1080 × 3 bytes
	=	6,220,800 bytes ≈ **6 MB**



File Signatures

- Many file formats begin with a recognizable pattern of bytes.
- This pattern is called a **file signature** or **magic number.**
- File signatures help software identify the file's actual format.

EXAMPLE SIGNATURES

File type	Example Signature (Hex)
PNG	89 50 4E 47 0D 0A 1A 0A
PDF-1.7	25 50 44 46 2D 31 2E 37
JPEG	FF D8 FF

Hex Dump: Understanding the Three Parts

- A **Hex dump** displays the bytes stored in a file.

1 Offset

Shows where each row begins in the file.

00000000 = offset 0

2 Hex Bytes

Shows the actual byte values in hexadecimal.

Each two-digit pair = one byte

3 ASCII Preview

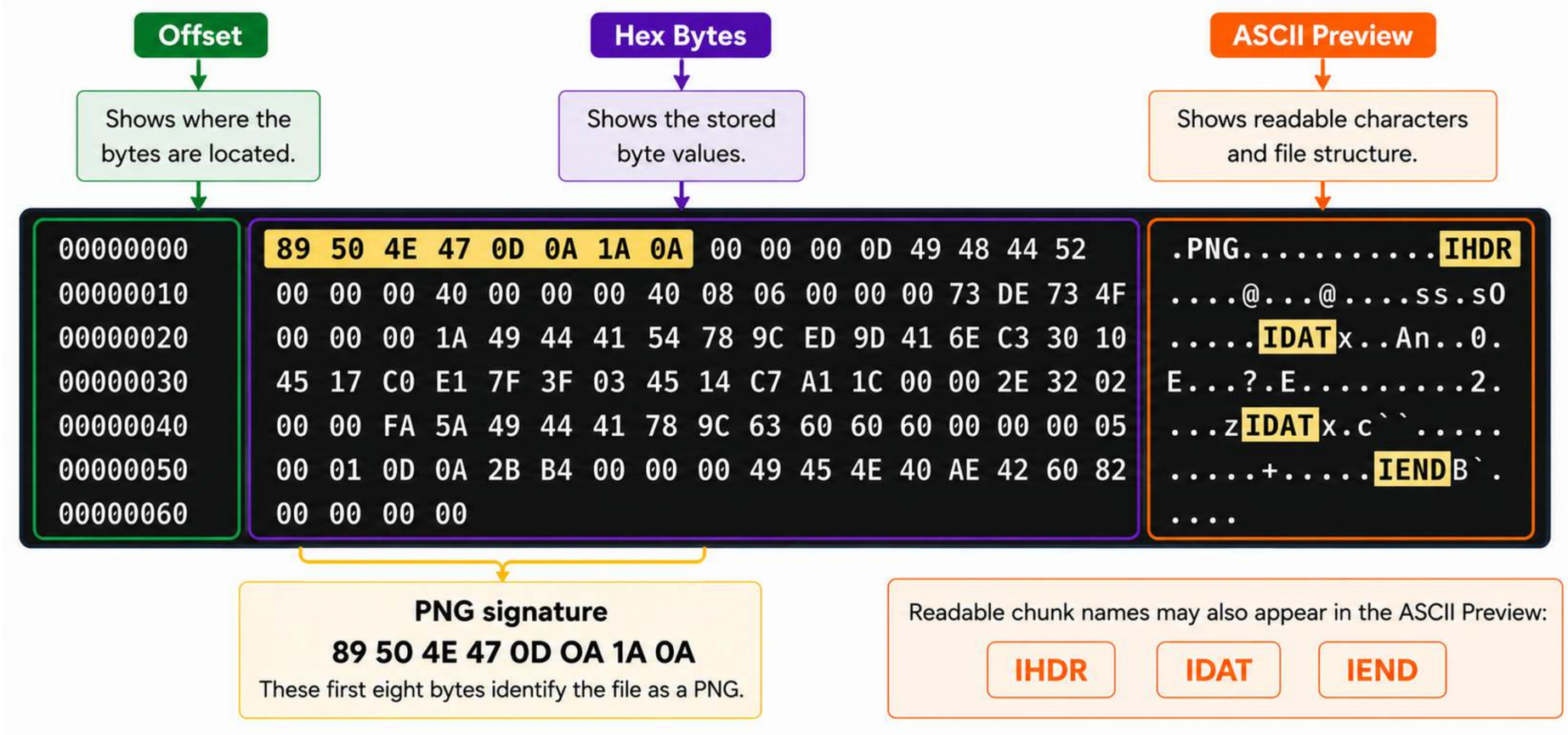
Shows the same bytes as readable characters.

Non printable bytes appear as dots.

OFFSET	HEX BYTES	ASCII PREVIEW
00000000	89 50 4E 47 0D 0A 1A 0A	.PNG....

Hex Dump: Inspecting a File in a Hex Editor

- A **hex editor** lets us view and **inspect the bytes** stored inside a file.



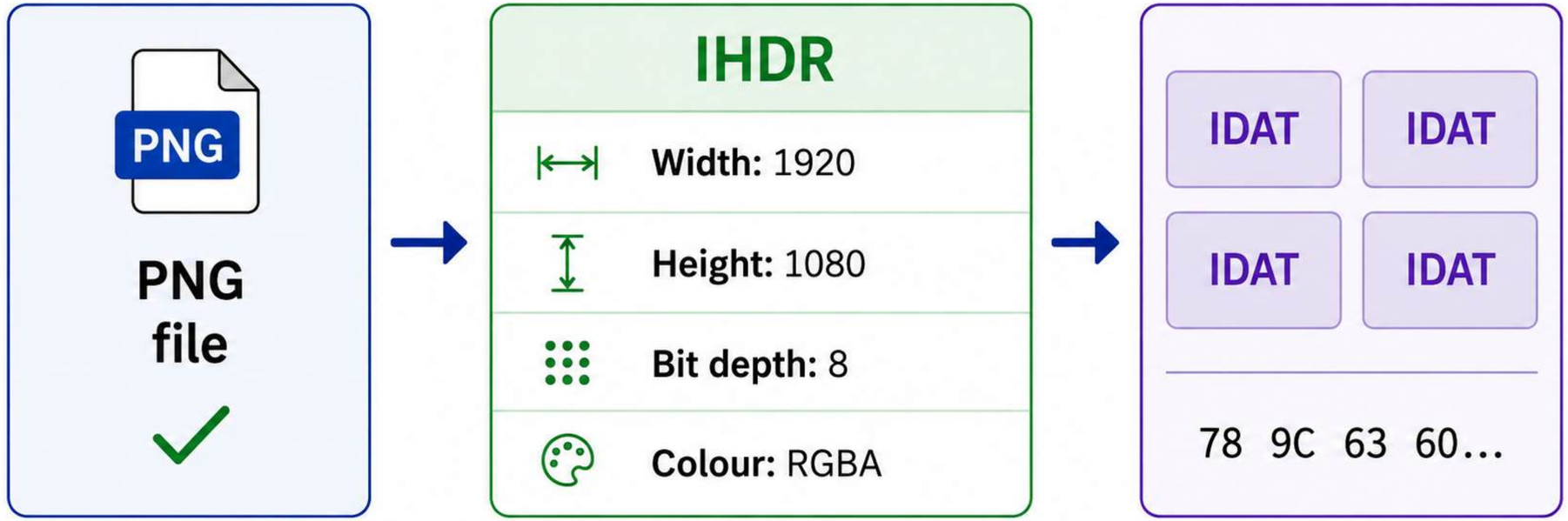
From Hex Bytes to PNG Chunks

Newton School of Technology

- A PNG stores different kinds of information in sections called chunks.
- Each chunk has a four-letter name stored as ASCII bytes:

HEX BYTES	ASCII NAME (Chunk Type Code)	MEANING
49 48 44 52	I H D R	**IHDR — Image Header** Stores the image width, height, bit depth and colour type.
49 44 41 54	I D A T	**IDAT — Image Data** Stores compressed image data.
49 45 4E 44	I E N D	**IEND — Image End** Marks the end of the PNG file.

From PNG Bytes to Pixels



Verify the Signature

Confirms it is a PNG.

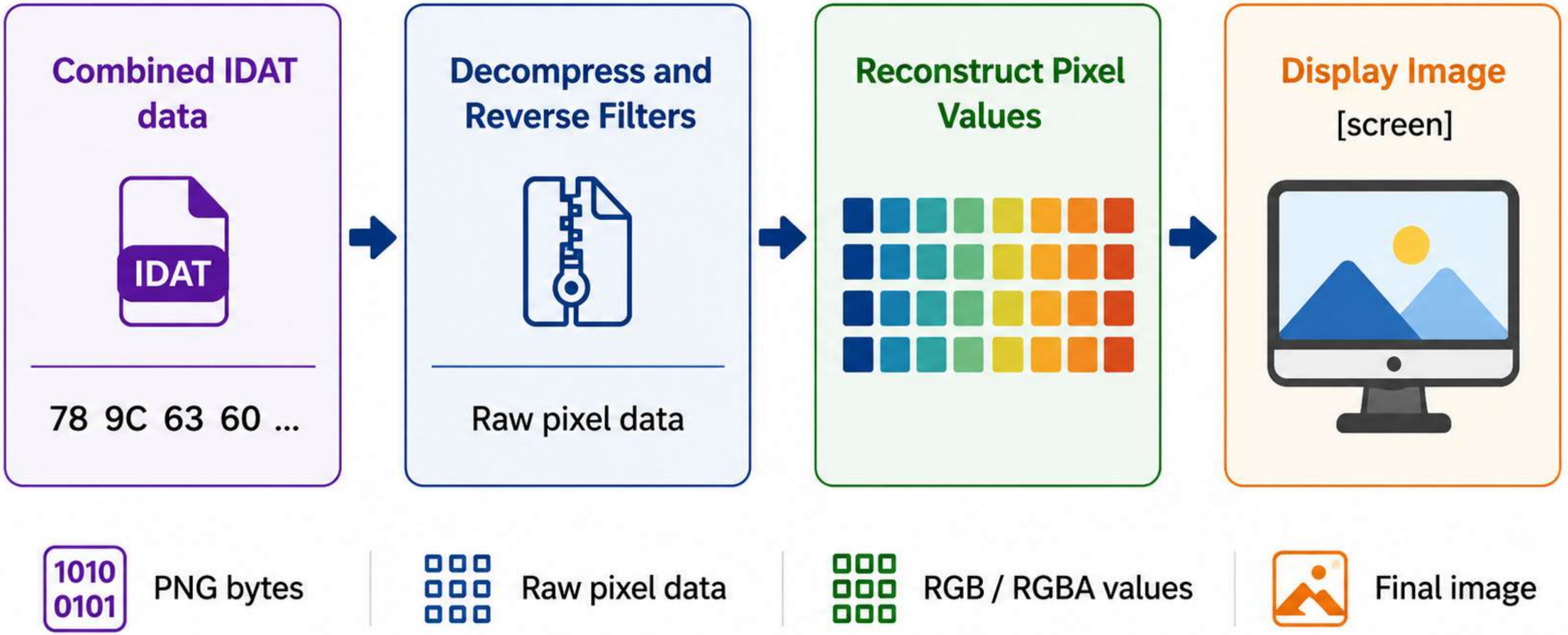
Read IHDR

Gets the image details.

Combine IDAT Data

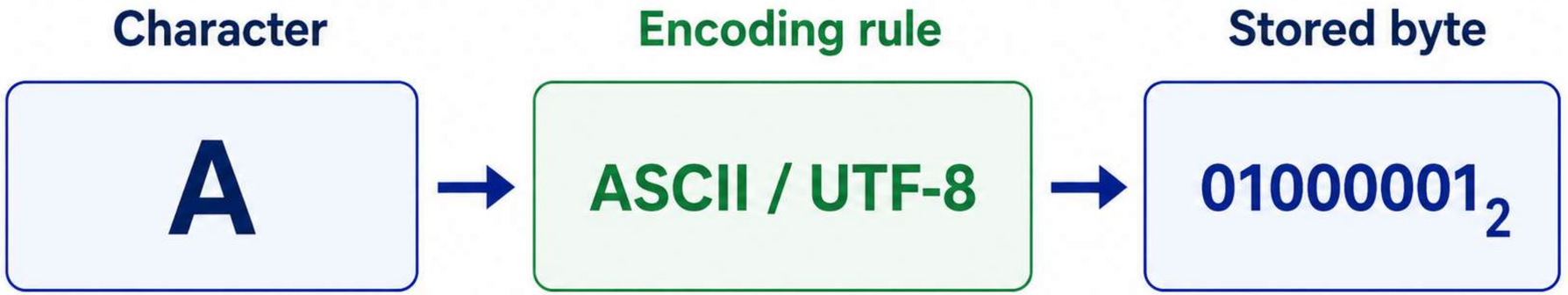
Combines IDAT data from all chunks in order.

From PNG Bytes to Pixels



Encoding vs Encryption

1. **Encoding: Characters become bytes**



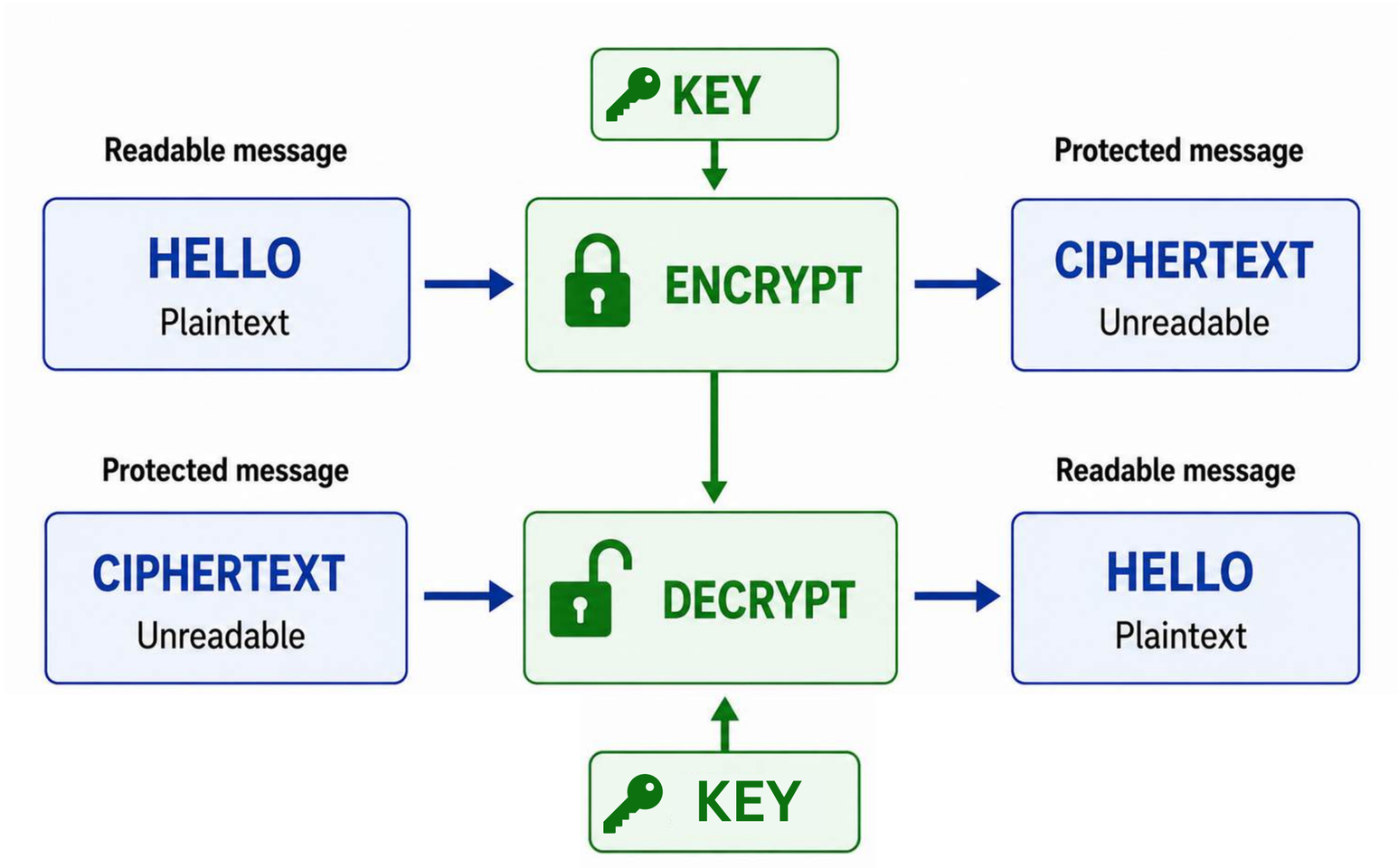
The same byte can be written in different ways:

65_{10} = 01000001_2 = 41_{16}

decimal binary hexadecimal

Encoding vs Encryption

2. **Encryption: Protects information**



Encoding vs Encryption

3. Key Differences at a Glance

Encoding	Encryption
Represents data	Protects data
Character → bytes	Plaintext → ciphertext
Uses a known rule	Uses a secret key
Anyone can read it with the rule	Only someone with the correct key can read it